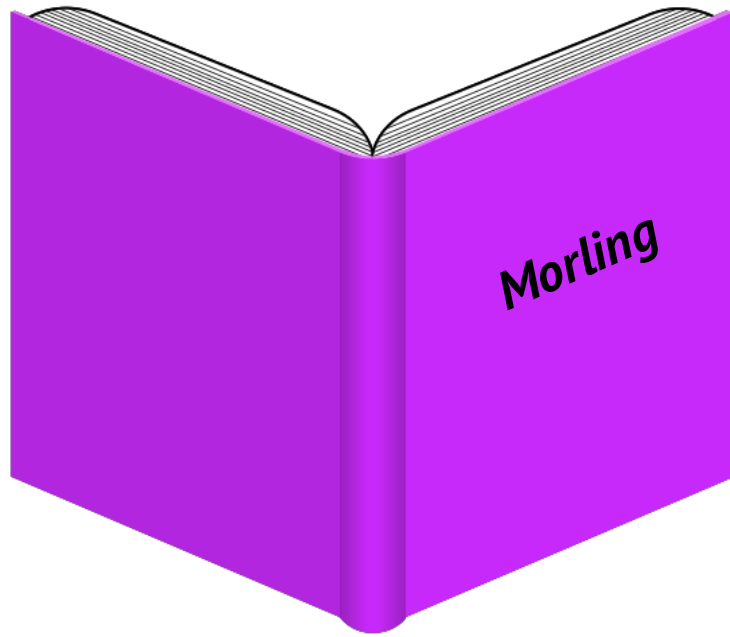




Research Methods and Statistics

Lecture 4: Variability in data + Association

Johnny van Doorn



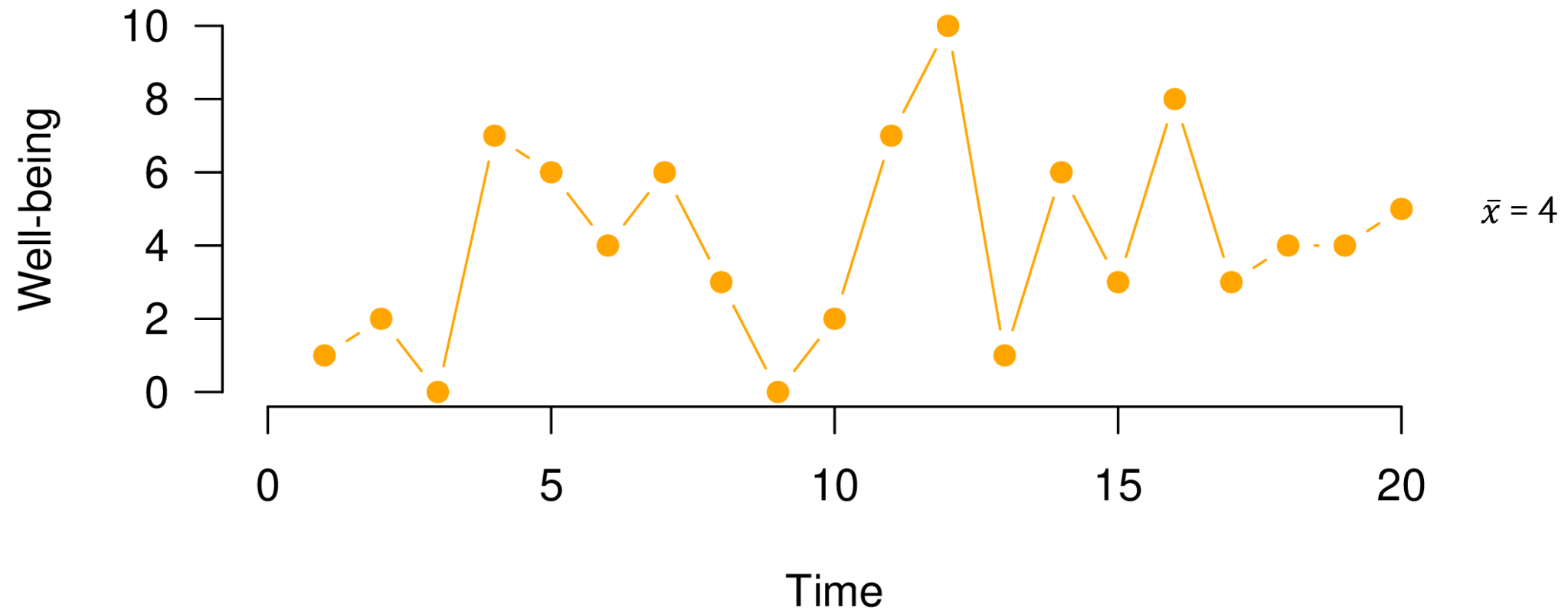
Pictures source: pixabay.org

Measuring Subjective Well-Being

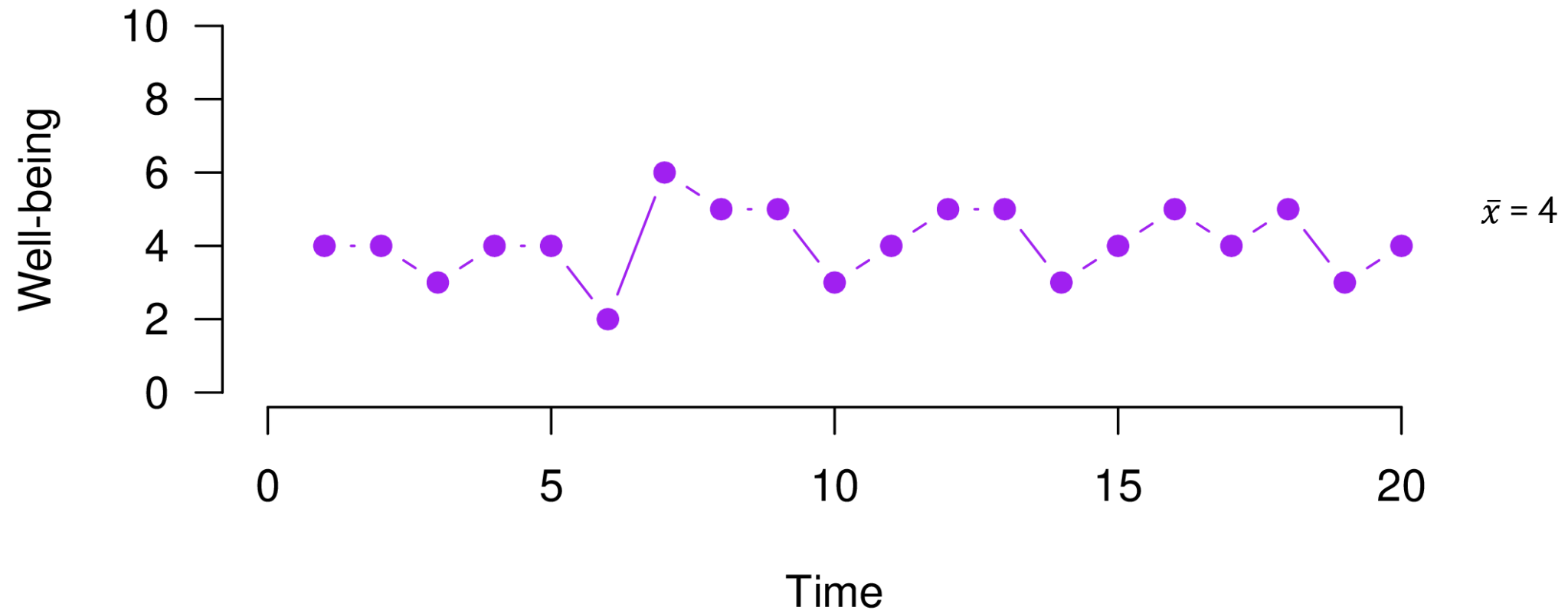
- Several measurements over time, on a discrete scale of 1-10
- 2 participants:
 - Person A: $\bar{x} = 4$
 - Person B: $\bar{x} = 4$



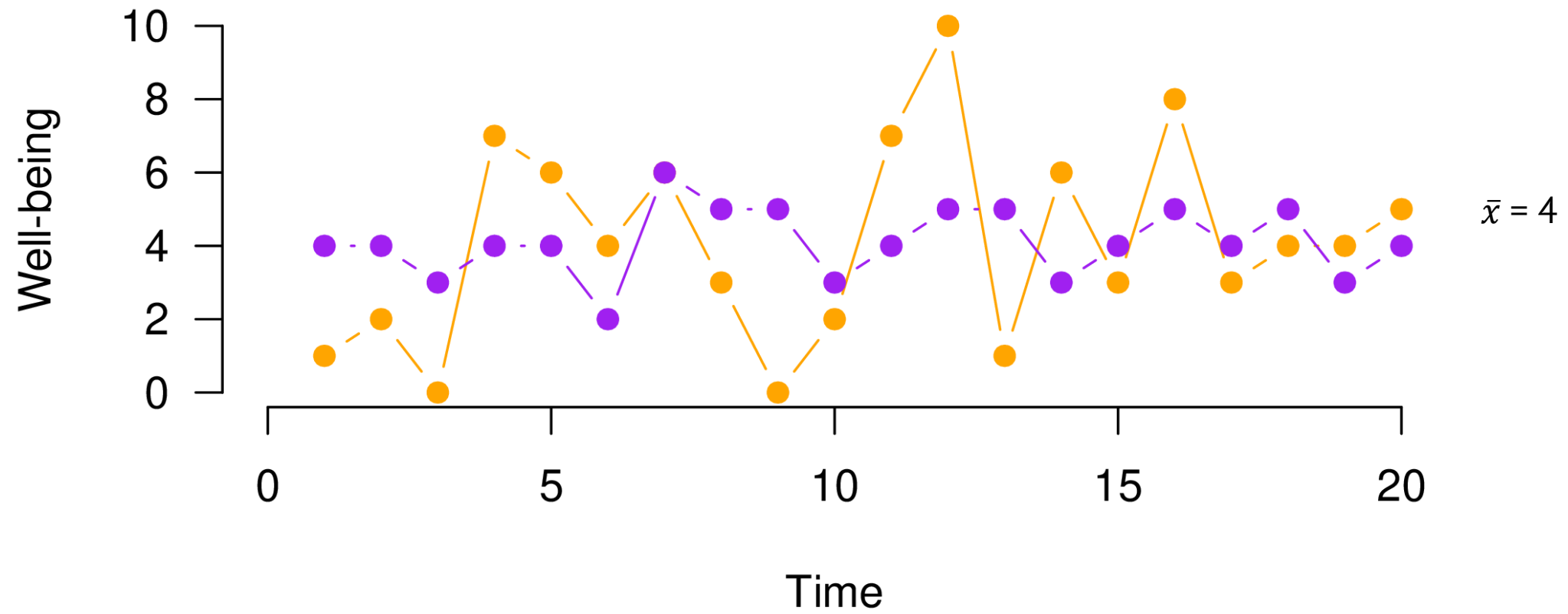
Person A



Person B



Person A & B



Today

1. Variability in the data

- Standard deviation and variance
- z-score
- Quartiles
- Boxplot

2. Associations

- Between two categorical variables
- Between two quantitative variables

3. Recap

- Next time
- Example exam question

How can we express/quantify variability of a variable?

How can we express/quantify an association between two variables?

Variability

- There are many ways to compute the *spread* of a distribution
 - Range
 - Absolute deviation
 - Variance + standard deviation
 - Inter Quartile Range
- Goal: Observe *variability*
 - Complements the location measures
 - To detect outliers

Standard deviation and variance

Variance: Average of the squared deviations

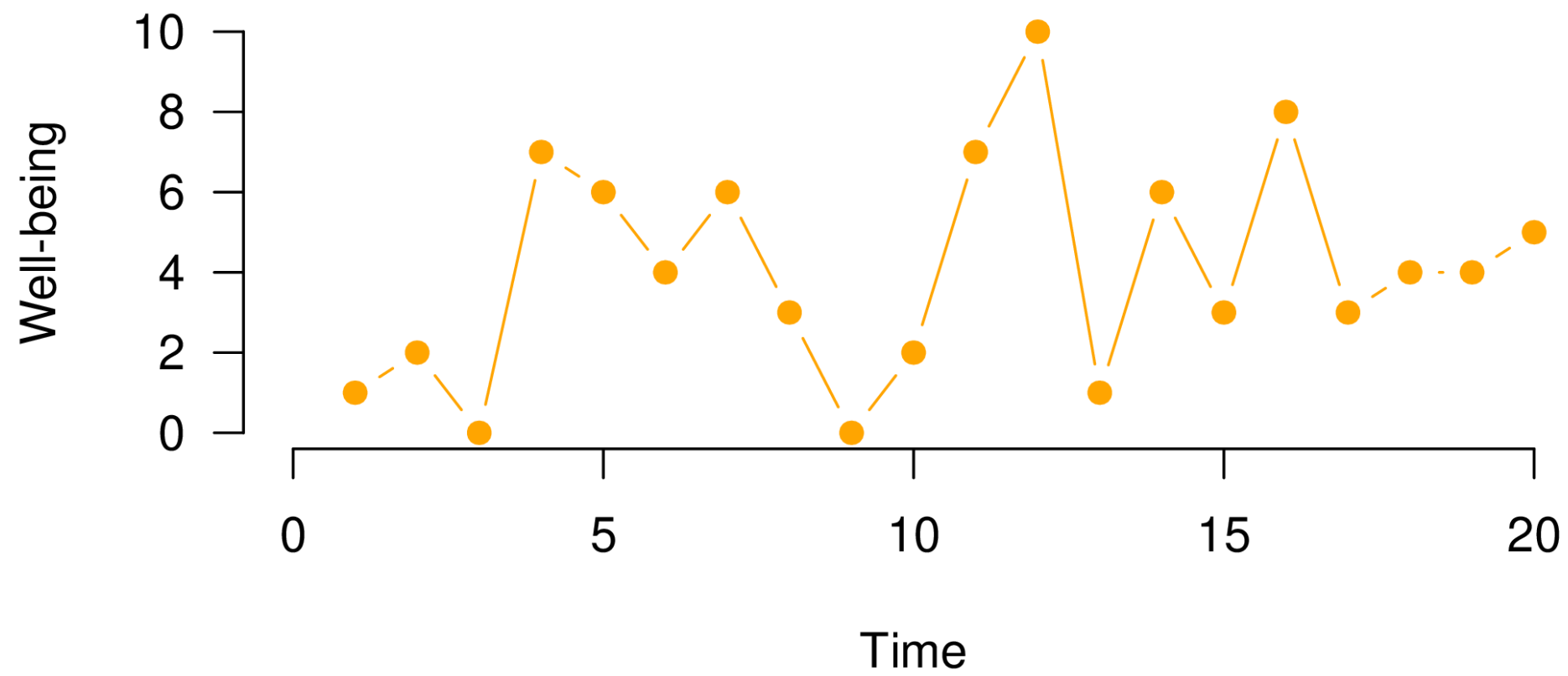
$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

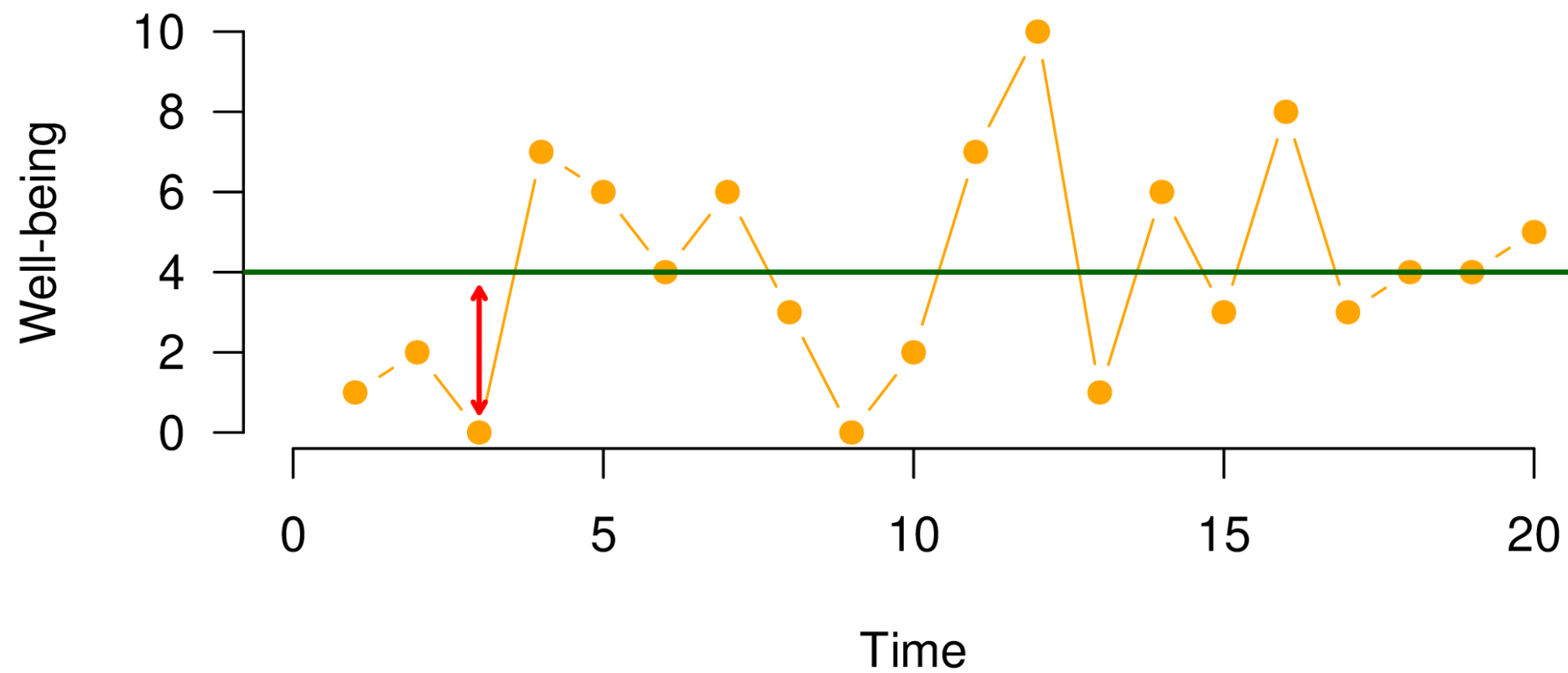
Standard deviation (sd): Square root of the variance

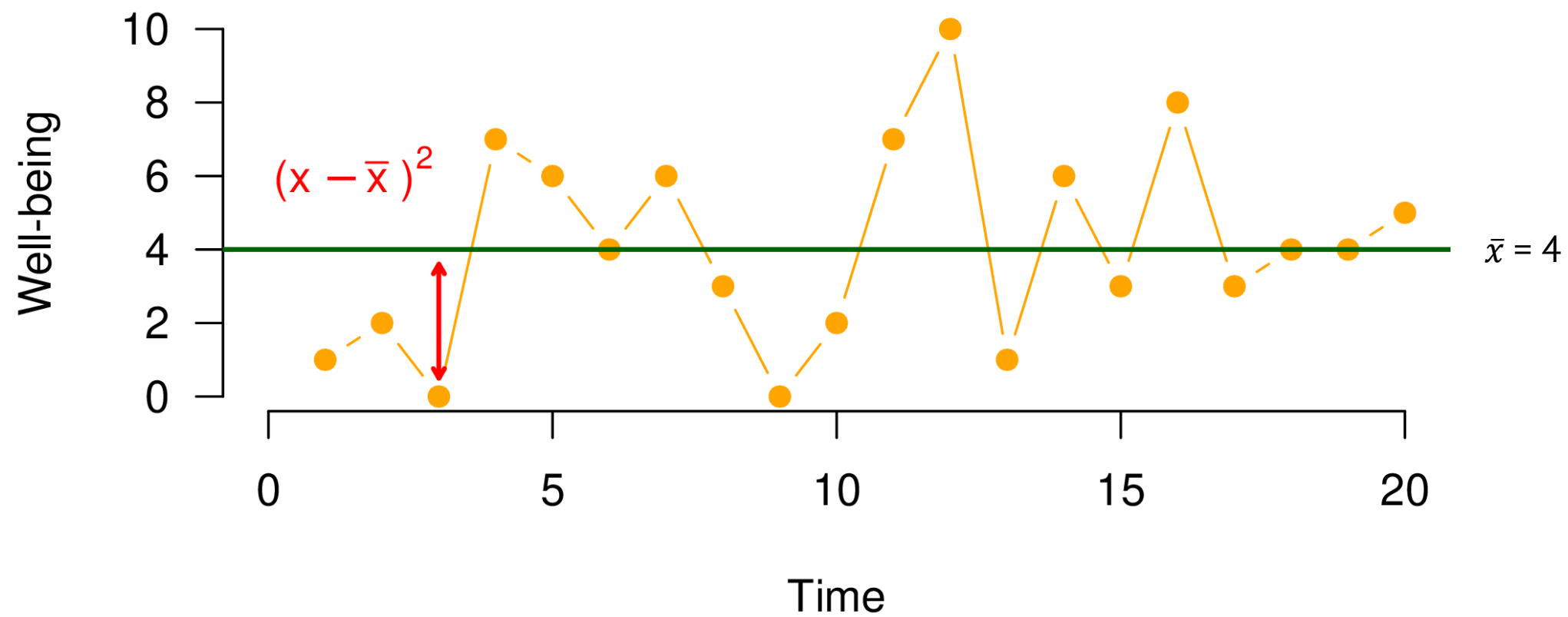
$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

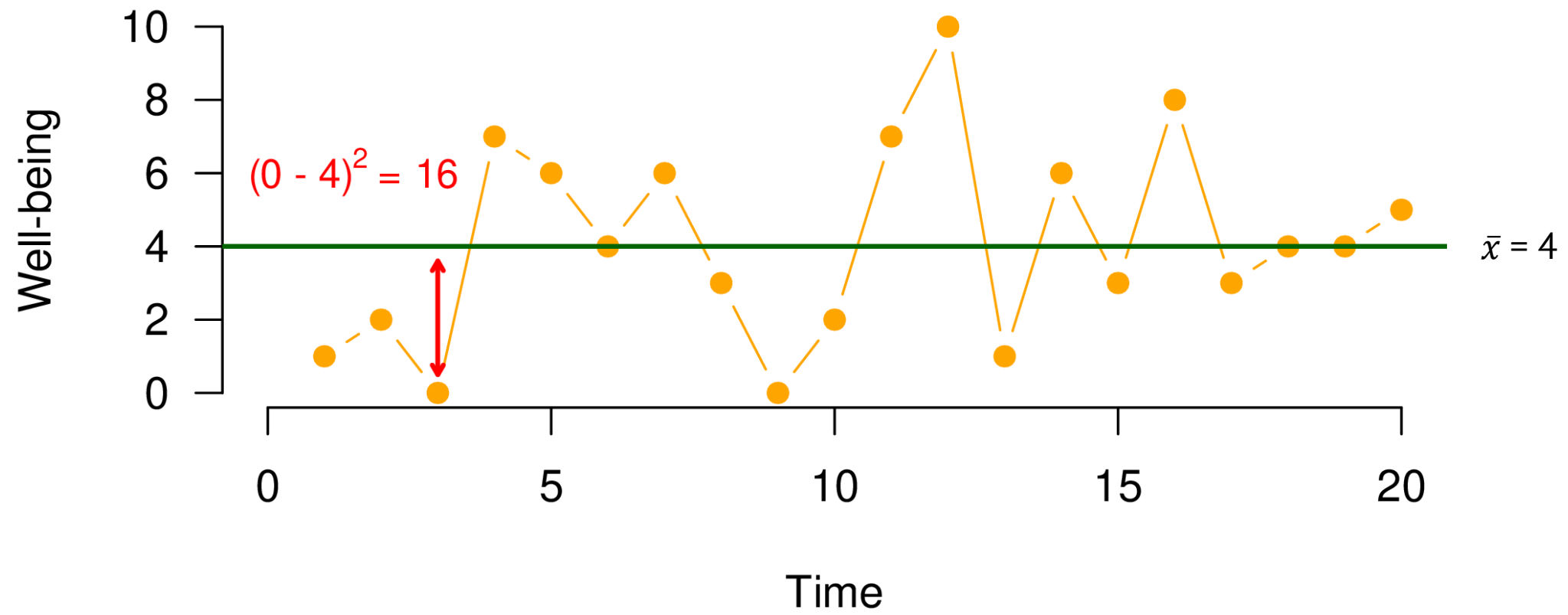
- Interpretation of s : A typical distance of an observation from the mean

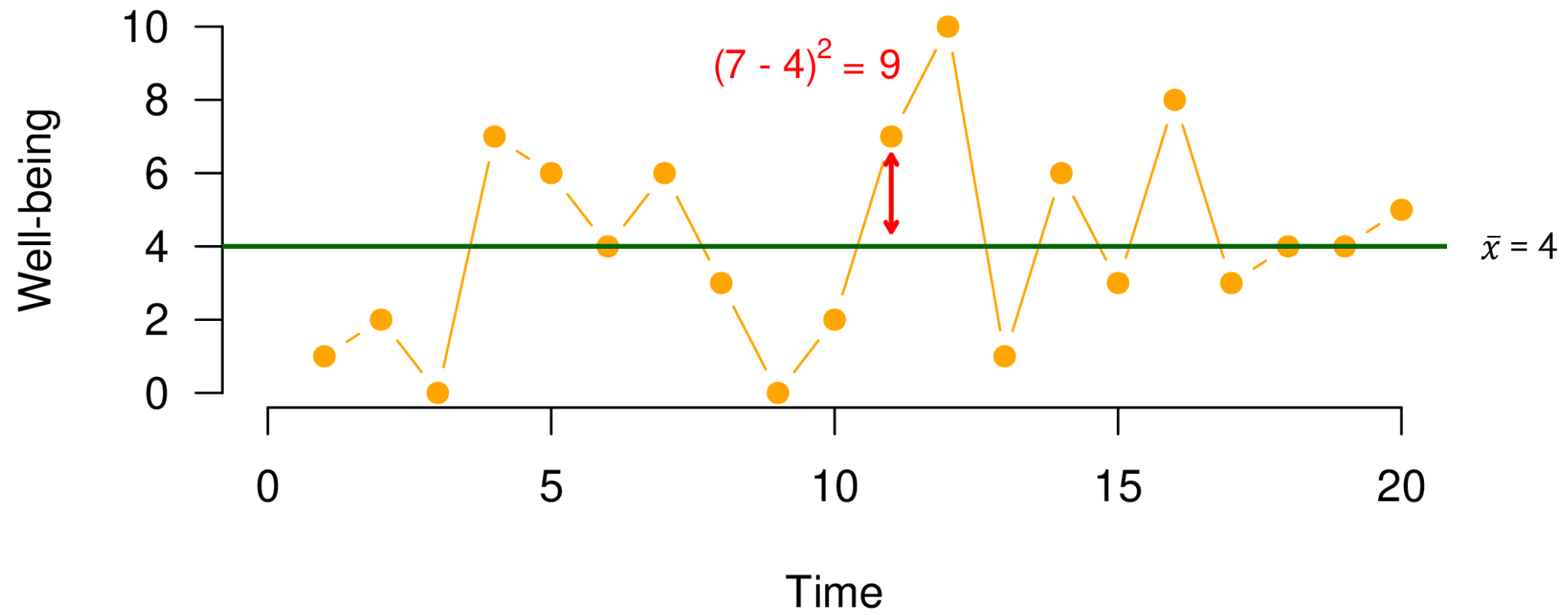
(Agresti, p. 91)

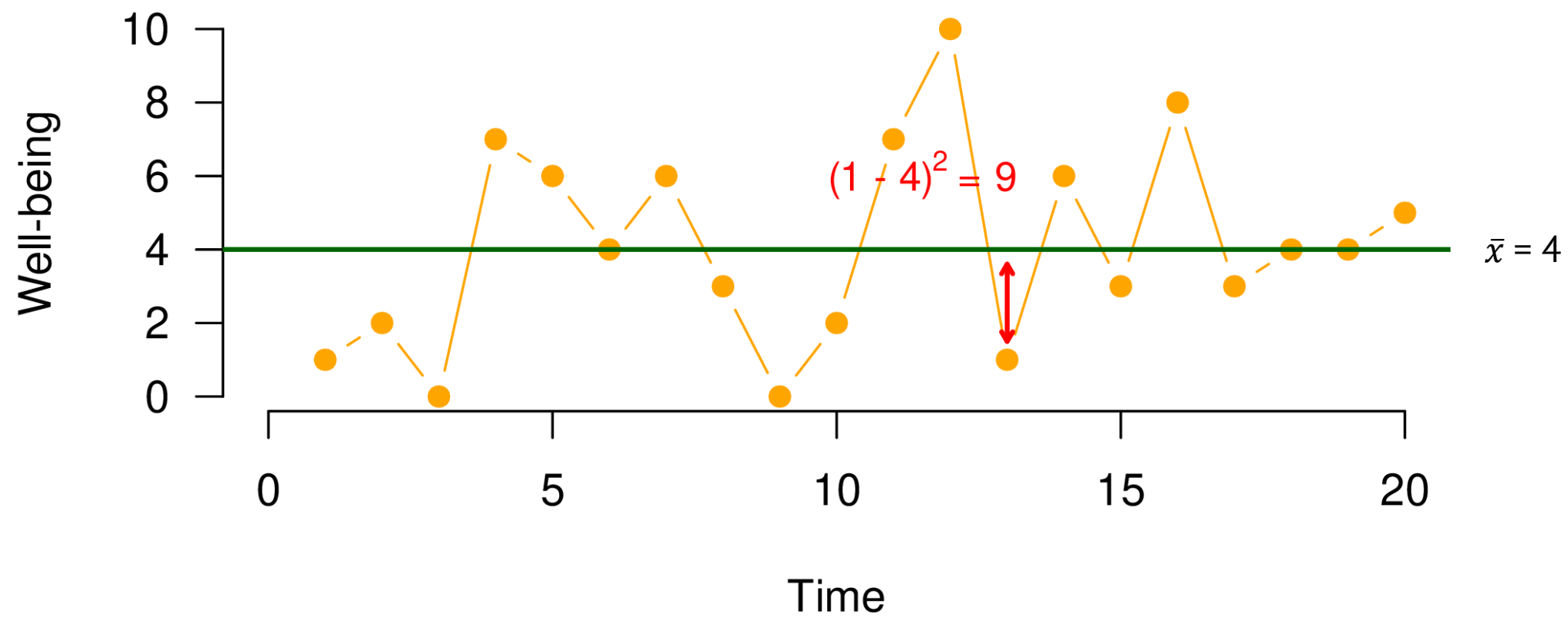


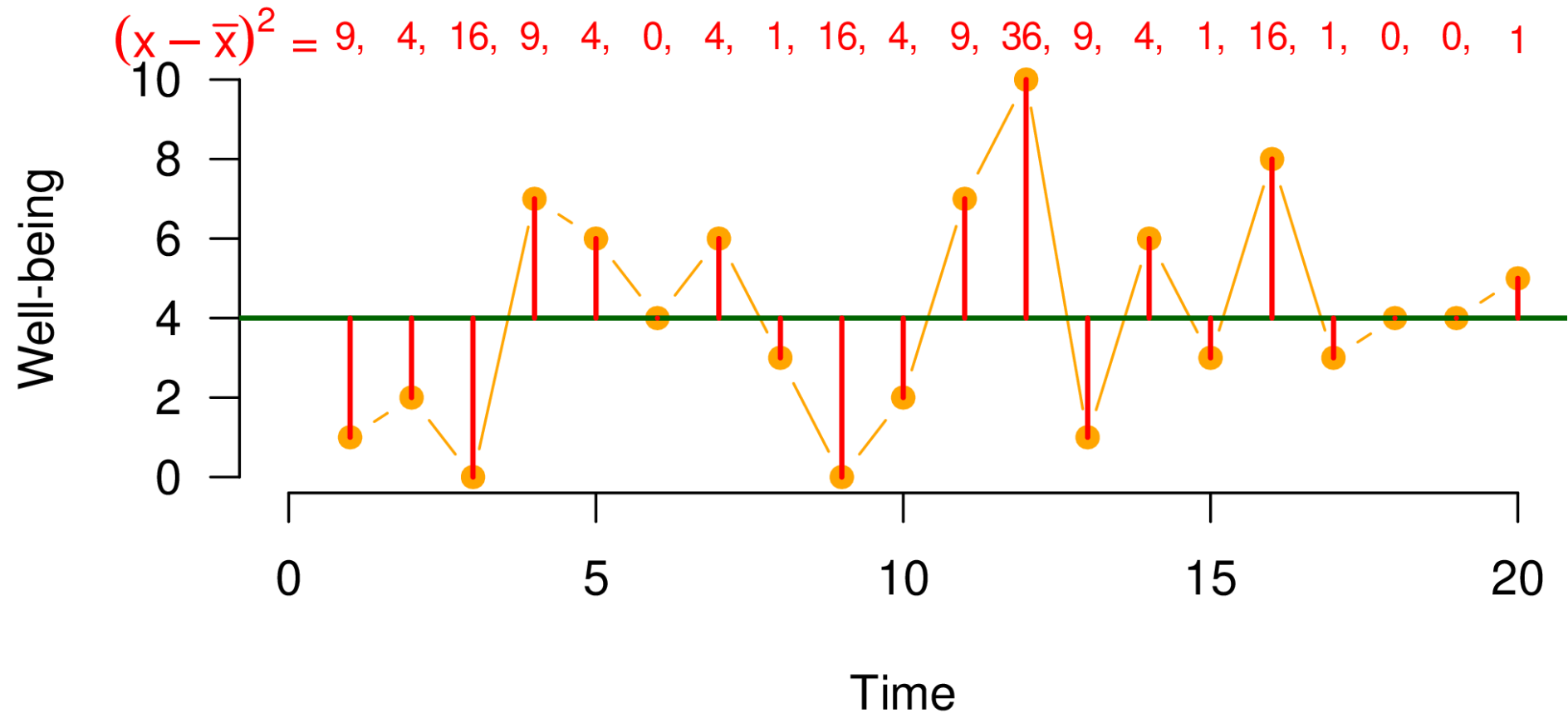


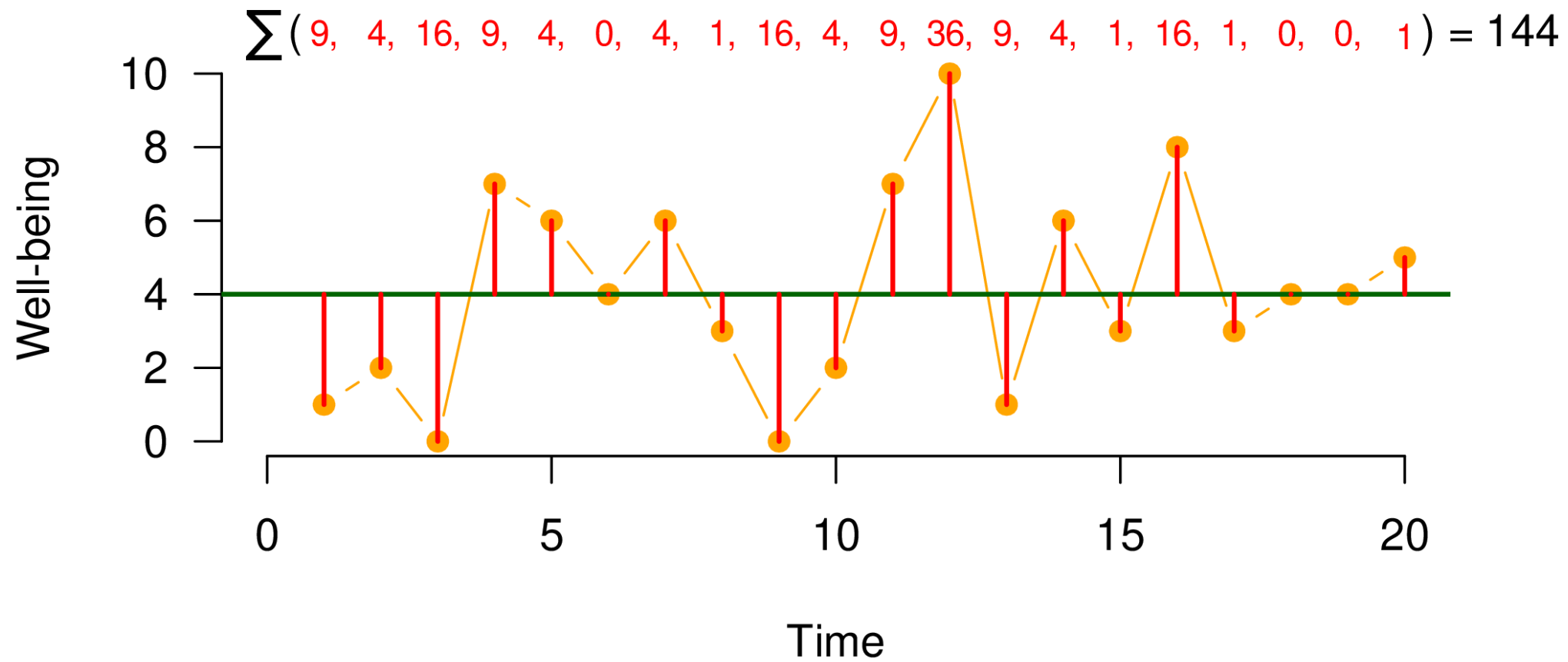


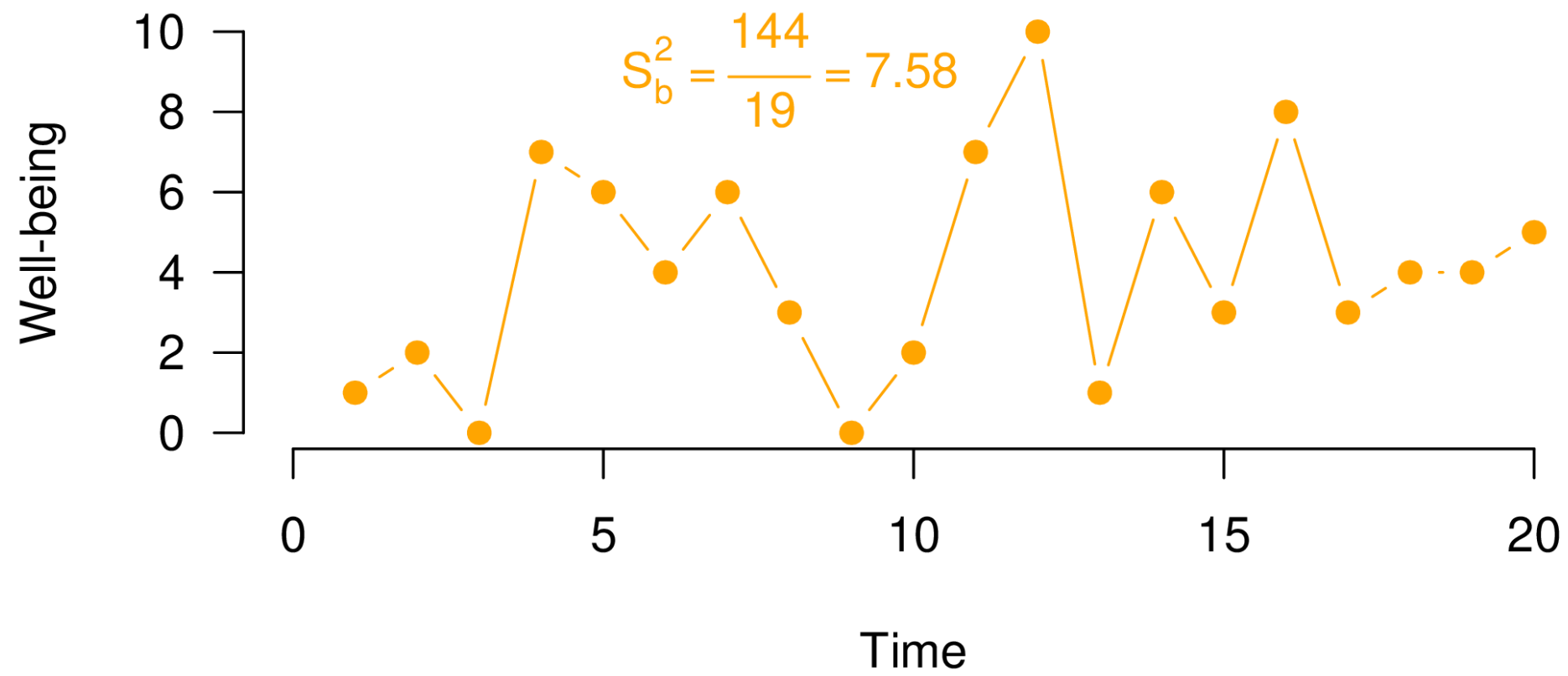


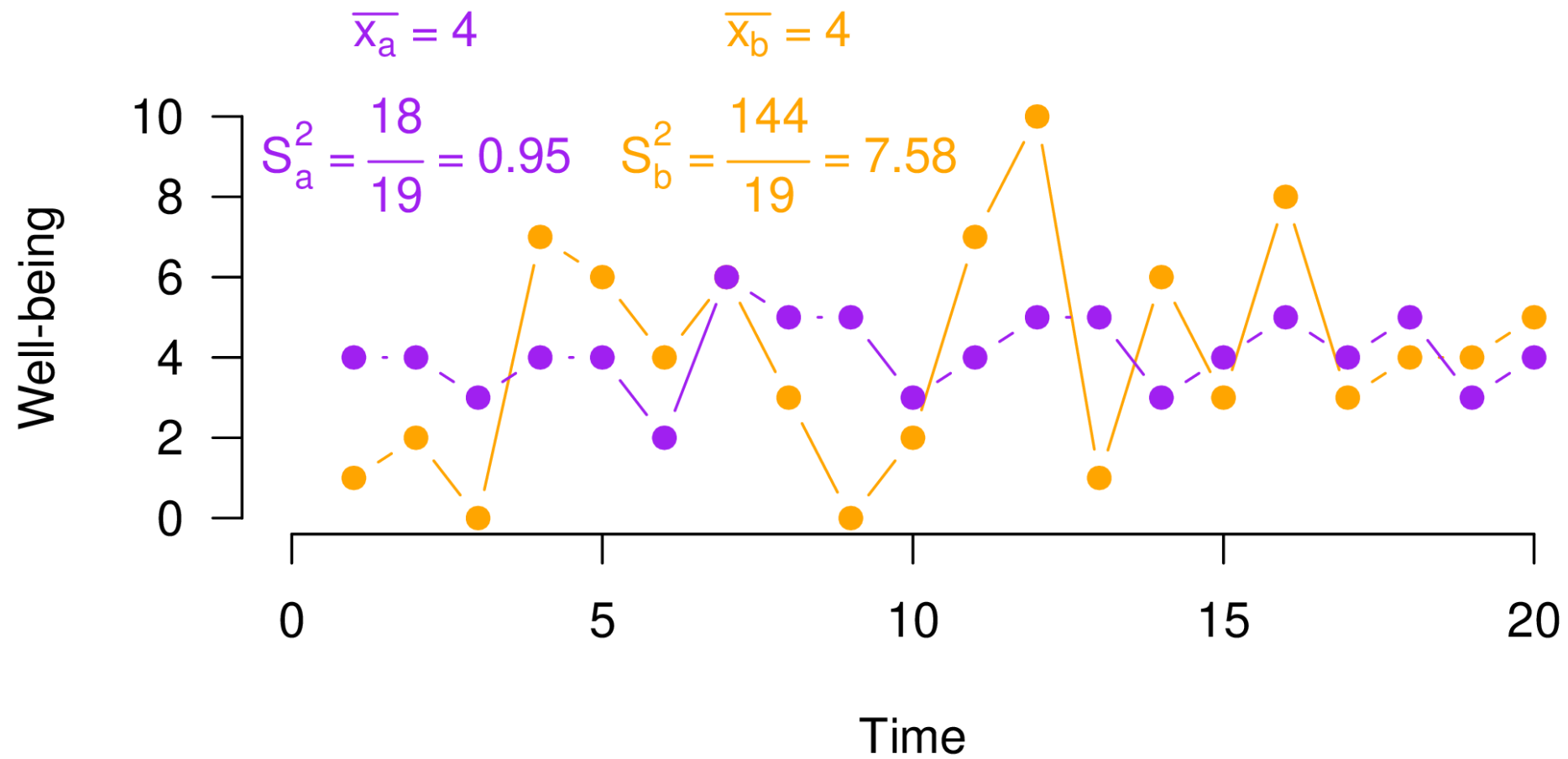












Standard deviation and variance

Variance: Average of the squared deviations

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Standard deviation: Square root of the variance

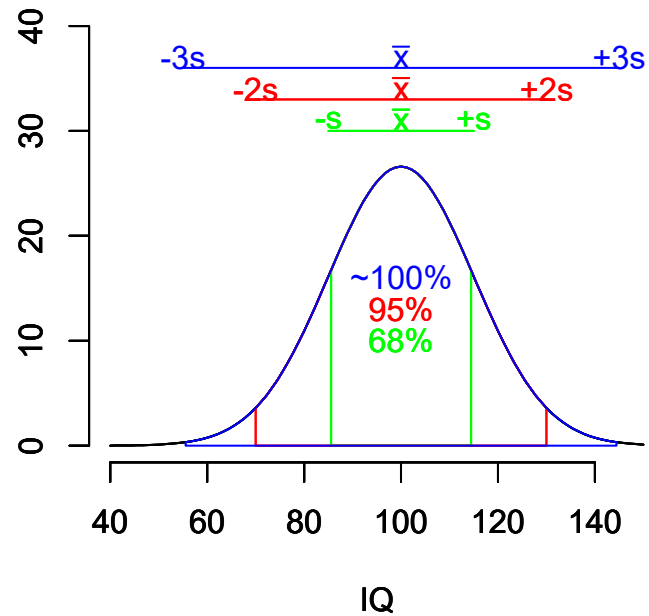
$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- Interpretation of s : A typical distance of an observation from the mean

The variance and sd express the same thing, just on a different scale ($s = \sqrt{s^2}$ the square root). By itself, the variance/sd does not mean much, but it can be informative to compare two variances/sd's

What can we now express?

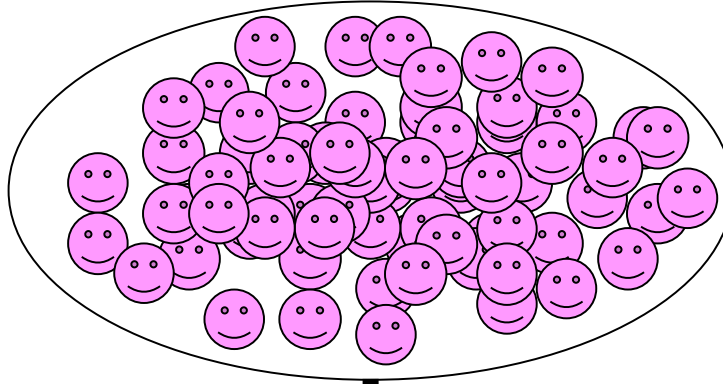
- *Proportion* of the data within a certain range
 - If the data is *Bell shaped*
 - *Empirical Rule*



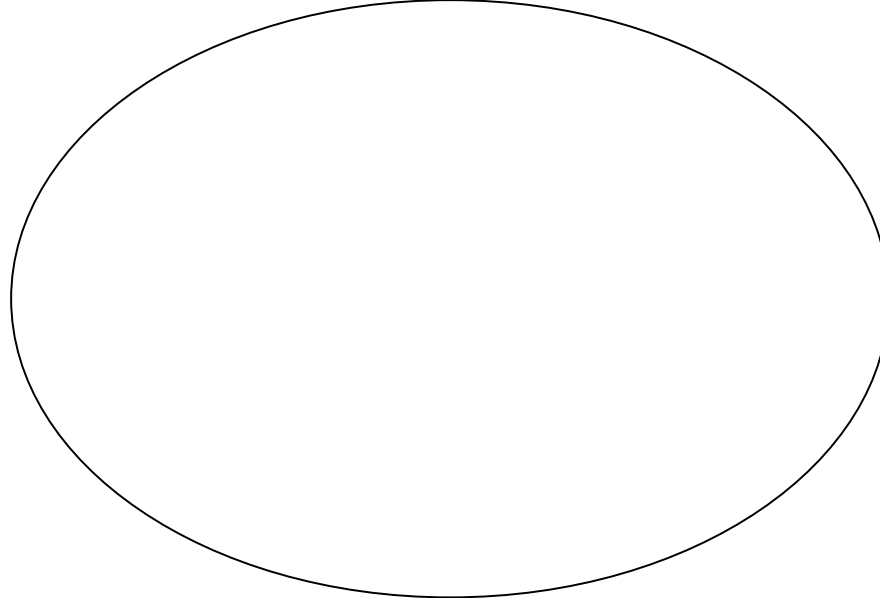
Symbols

	Sample (statistics, in Latin)	Population (parameters, in Greek)
Location: mean	$\bar{x} = \frac{\sum x}{n}$	μ
Variability: variance	$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$	σ^2
Variability: standard deviation	$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$	σ

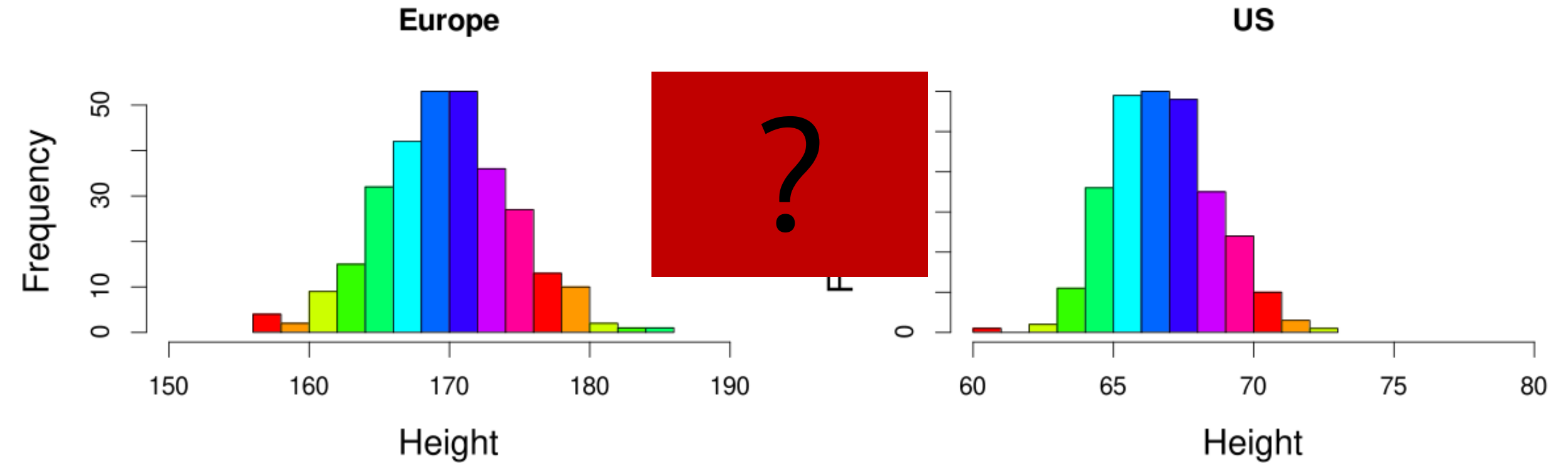
Population
 (μ, σ)



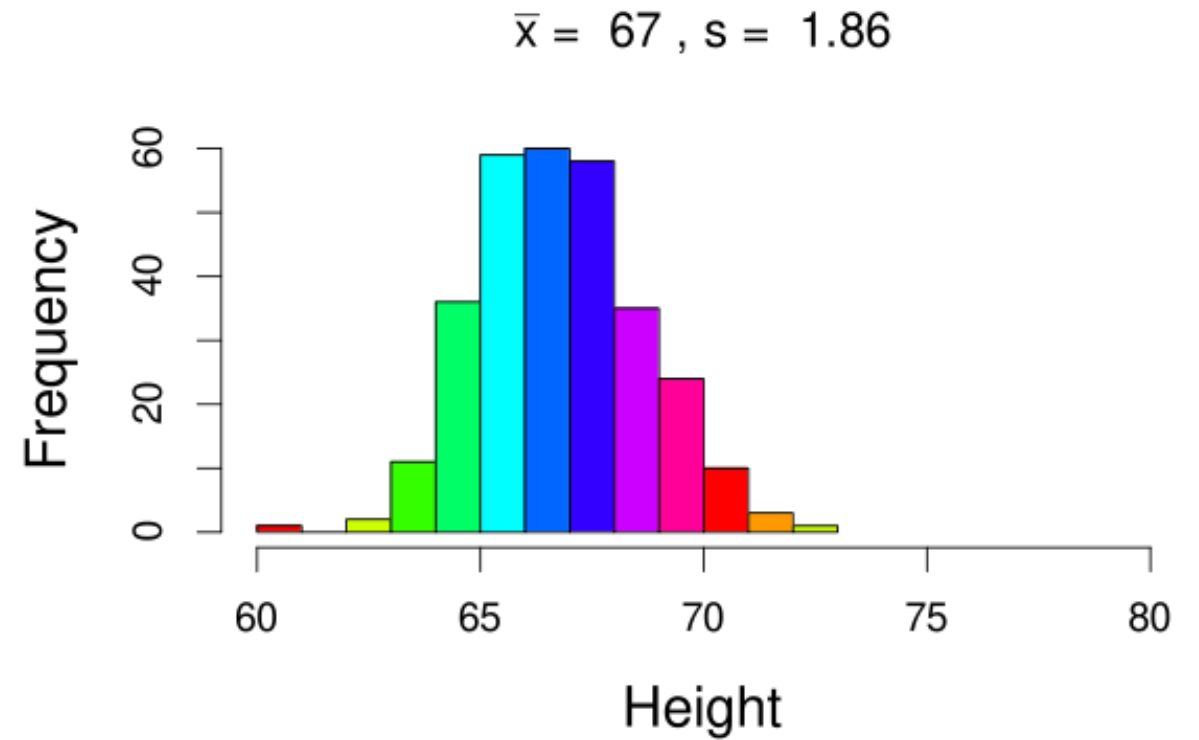
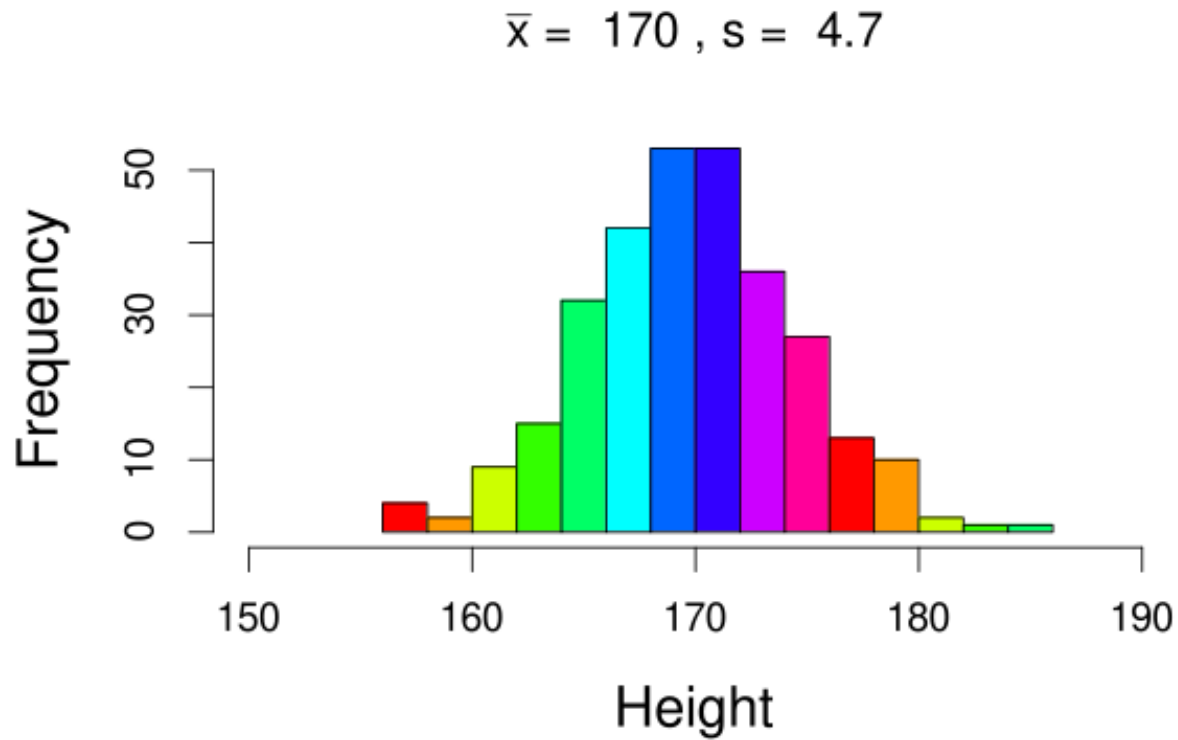
Sample
 $(\bar{x}, s)$



Measuring Height Across Cultures



Measuring Height Across Cultures



Measuring Height Across Cultures

$$\bar{x} = 170, s = 4.7$$



Source: <https://pixabay.com/>

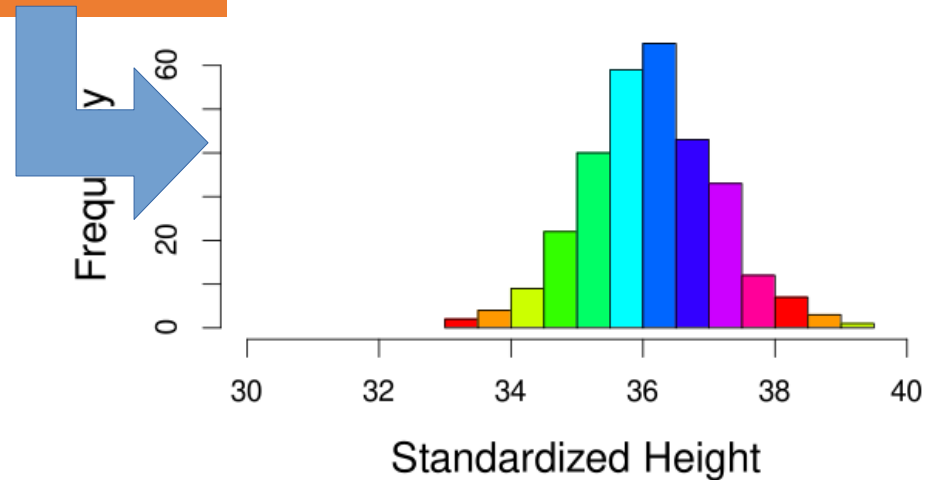
$$\bar{x} = 67, s = 1.86$$



Measuring Height Across Cultures

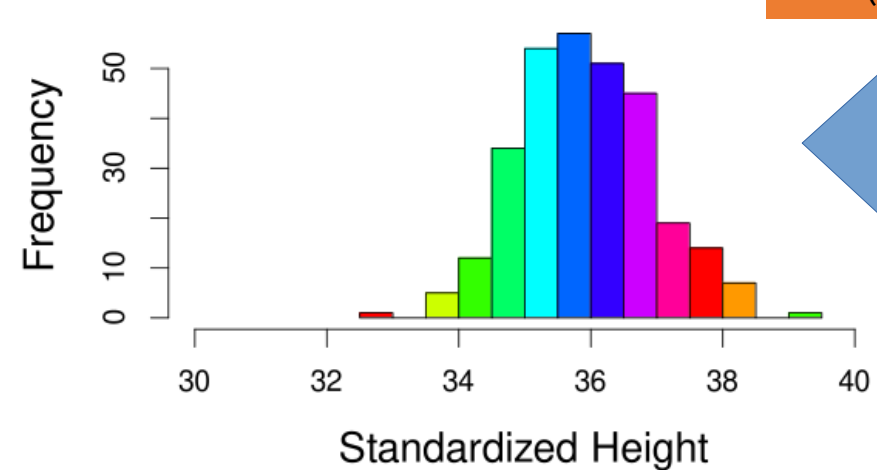
We can use the standard deviation to **standardize** the scores, such that the original measurement scales do not matter:

We divided all European heights by 4.7 (= sd)



$\bar{x} = 36, s = 1$

We divided all American heights by 1.86 (= sd)



Measuring Height Across Cultures

We can use the standard deviation to **standardize** the scores, such that the original measurement scales do not matter:

We divided all European heights by 4.7 (= sd)

Frequency



We divided all American heights by 6 (= sd)

Frequency

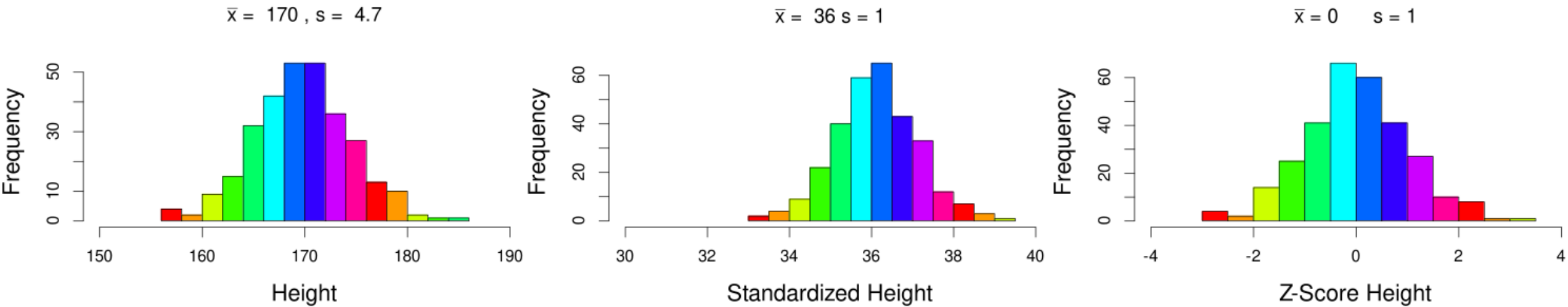


Z-score is a way to also standardize the mean

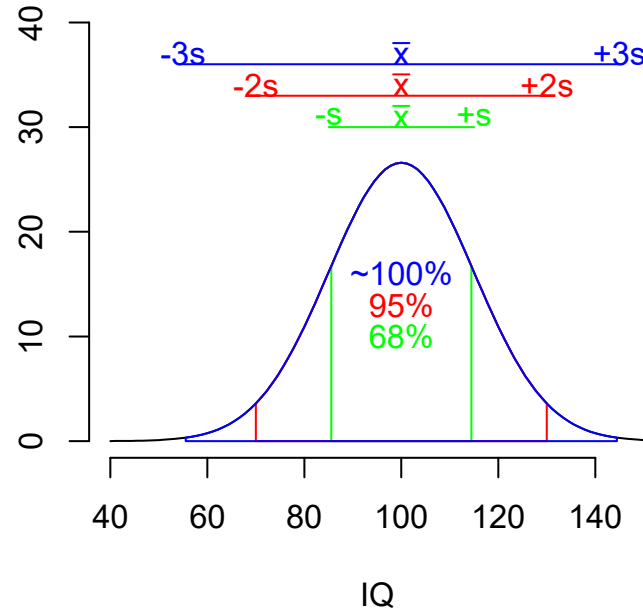
- You can recompute all observations to a z-score
- Mean is always 0
- Standard deviation is always 1

$$\text{z-score:} \quad z = \frac{\text{observation} - \text{mean}}{\text{standard deviation}}$$

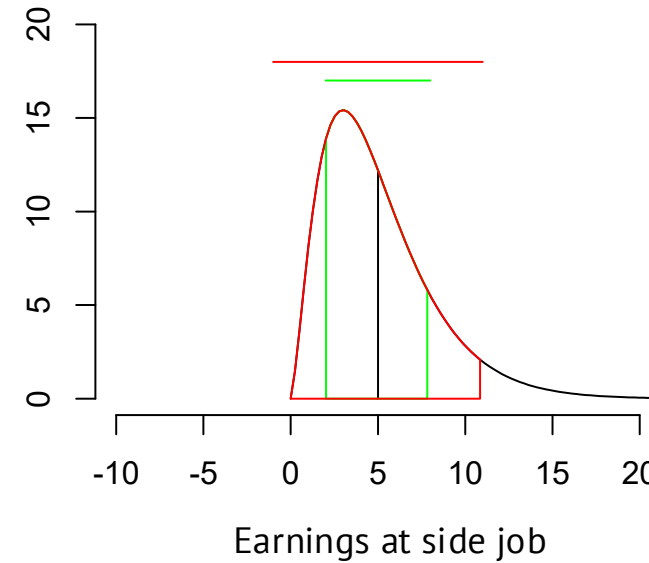
Z-score is a way to also standardize the location



Empirical rule only applies to Bell shaped distributions



Symmetric: Equal proportion left and right of the mean



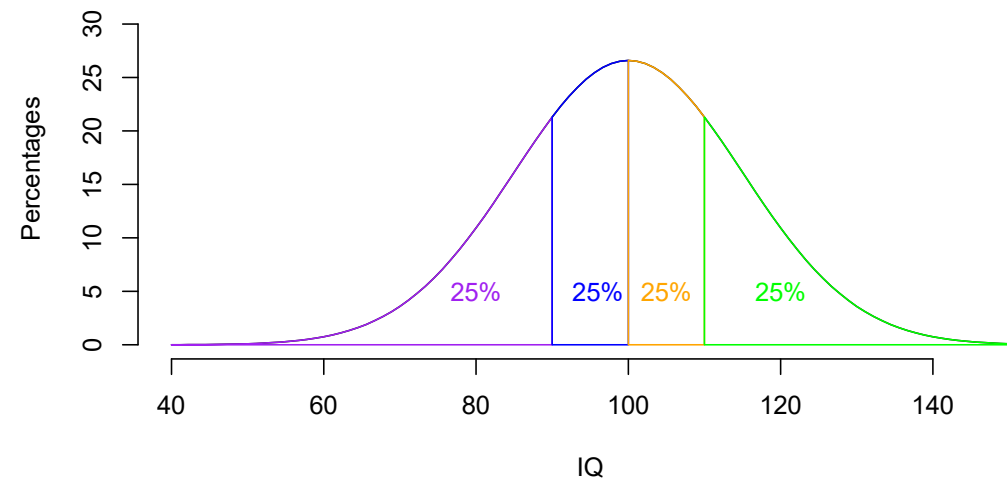
(Right) skewed: Higher proportion right than left

Other solution: Quartiles

- Proportion of the data within a certain range



Q1 = 242: 25% earns less than 242 euro
Q2 = 473: 50% earns less than 473 euro
= Median
Q3 = 821: 75% earns less than 821 euro



Agresti, p. 99 explains how to find the quartiles
→ You do not need to find these yourself, but understand them

Box plot

- IQR: Interquartile range
 - $IQR = Q3 - Q1 = 900 - 200 = 700$
- Whiskers
 - $Q1 - 1.5 \times IQR$ OR minimum
 - $250 - 1.5 \times 700 = -800$
 - $Q3 + 1.5 \times IQR$ OR maximum
 - $800 + 1.5 \times 700 = 1850$
 - Take the option that is closest to the box
 - Left whisker: minimum
 - Right whisker: $Q3 + 1.5 \times IQR$
 - Outliers?



Today

1. Variability in the data

- Standard deviation and variance
- z-score
- Quartiles
- Boxplot

2. Associations

- Between two categorical variables
- Between two quantitative variables

3. Recap

- Next time
- Example exam question

Association

Association: There is an association if a particular value of one variable is more likely to occur with certain values of the other variable

- One variable *depends on* the other variable
- Two variable types:

Response variable: The variable for which you want to explain/predict the outcome

Explanatory variable: The variable that you use to predict

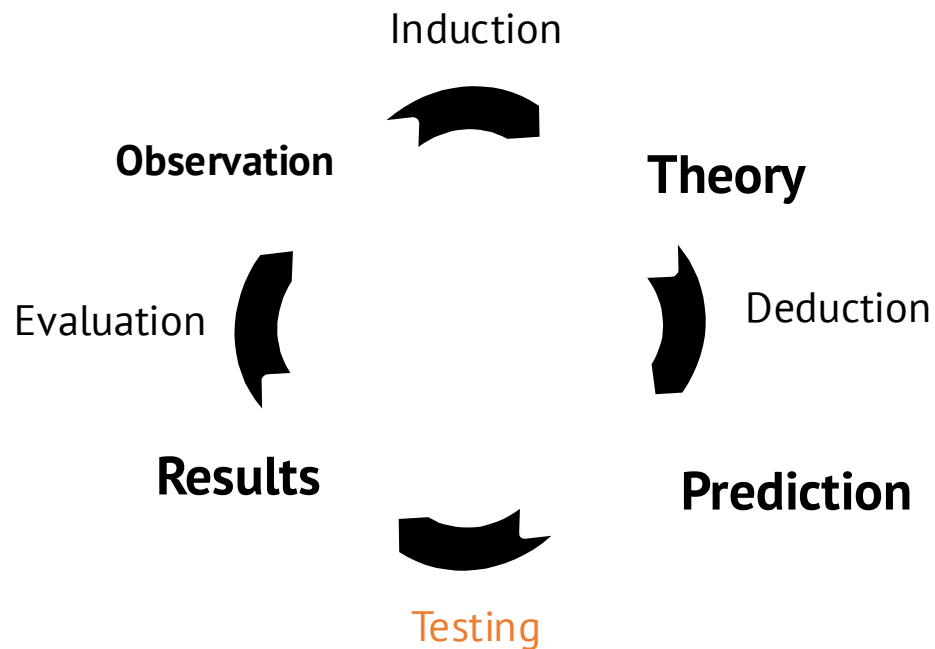
- Note different terminology:
 - Response variable → dependent variable
 - Explanatory variable → independent variable, predictor variable

Response or explanatory variable?

- Are older students generally happier?
 - “Age student”?
 - Explanatory
 - “Subjective well-being score”?
 - Response
- Does sitting in a dark room enhance cognitive ability?
 - “Reaction time”
 - Response
 - “Room illumination”
 - Explanatory

Role of variables in empirical cycle

Operationalize: determine how you will measure the conceptual variables from the prediction



Role of variables in empirical cycle

		Explanatory variable	
Response variable		Quantitative	Categorical
	Quantitative	Correlation Regression	t-test ANOVA
	Categorical	Logistic regression	Contingency table

Today

1. Variability in the data
 - Standard deviation and variance
 - z-score
 - Quartiles
 - Boxplot
2. Associations
 - **Between two categorical variables**
 - Between two quantitative variables
3. Closing
 - Next time
 - Example exam question

Association between two categorical variables



One flew over the Cuckoo's nest (1975)
source: https://i.ytimg.com/vi/d_mASr1djMM/maxresdefault.jpg

Association between two categorical variables

- Research question: Do people that are treated with electroshock therapy recover from a psychotic disorder?
- Both the response and explanatory variables are categorical (yes/no)
 - Contingency table!

		Explanatory variable	
		Quantitative	Categorical
Response variable	Quantitative	Correlation Regression	t-test ANOVA
	Categorical	Logistic regression	Contingency table

Contingency Table

Counts

Recovered?

Treatment?

	No	Yes	Total
No	21	19	40
Yes	13	27	40
Total	34	46	80(= n)

(Agresti, p. 137)

Proportions

Recovered?

Treatment?

	No	Yes	Total
No	0.26	0.24	0.5
Yes	0.16	0.34	0.5
Total	0.42	0.58	80(= n)

(e.g., $21 / 80 = 0.26$)

Conditional Proportions

Conditional proportion: The proportion of the response variable, for *one level* of the explanatory variable

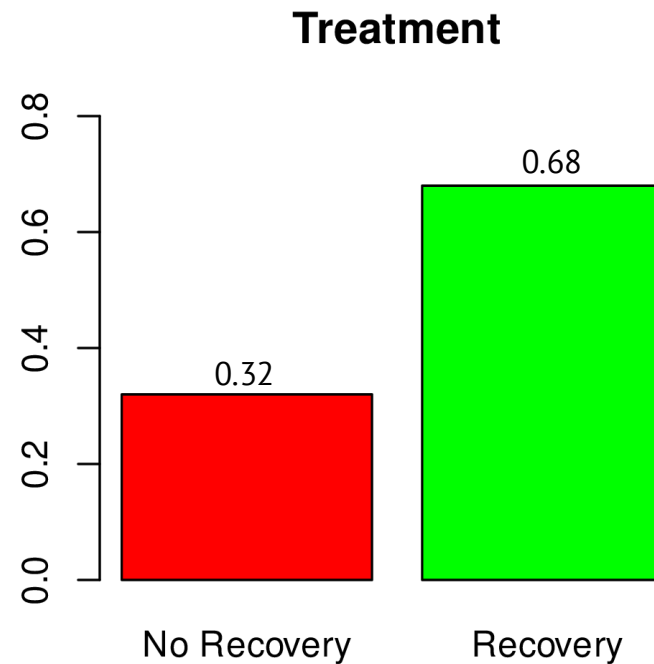
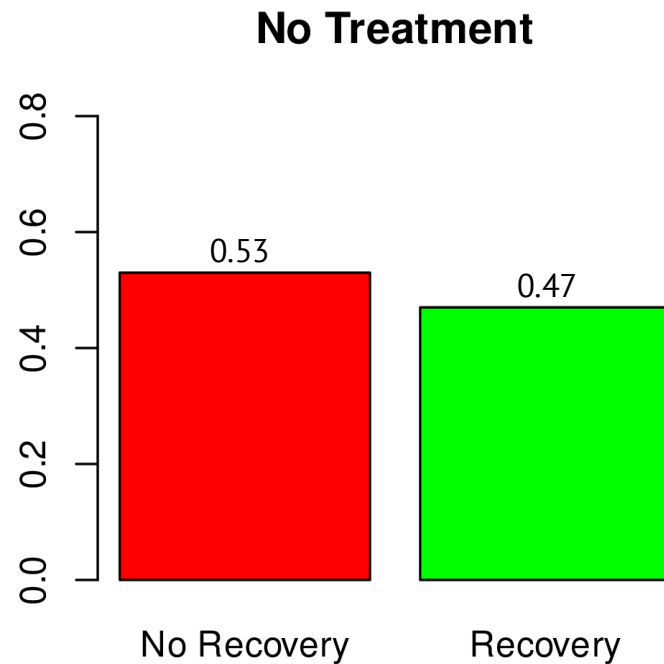
Treatment?	Recovered?		
	No	Yes	Total
	No	21 19	40
	Yes	13 27	40
Total		34 46	80(= n)

Treatment?	Recovered?		
	No	Yes	Total
	No	0.53 0.47	1
	Yes	0.32 0.68	1
Total			80(= n)

(e.g., $13 / 40 = 0.32$)

Conditional Proportions

Conditional proportion: The proportion of the response variable, for *one level* of the explanatory variable



Treatment?	Recovered?		
	No	Yes	Total
	0.53	0.47	1
	0.32	0.68	1
Total			80(= n)

What can we now express?

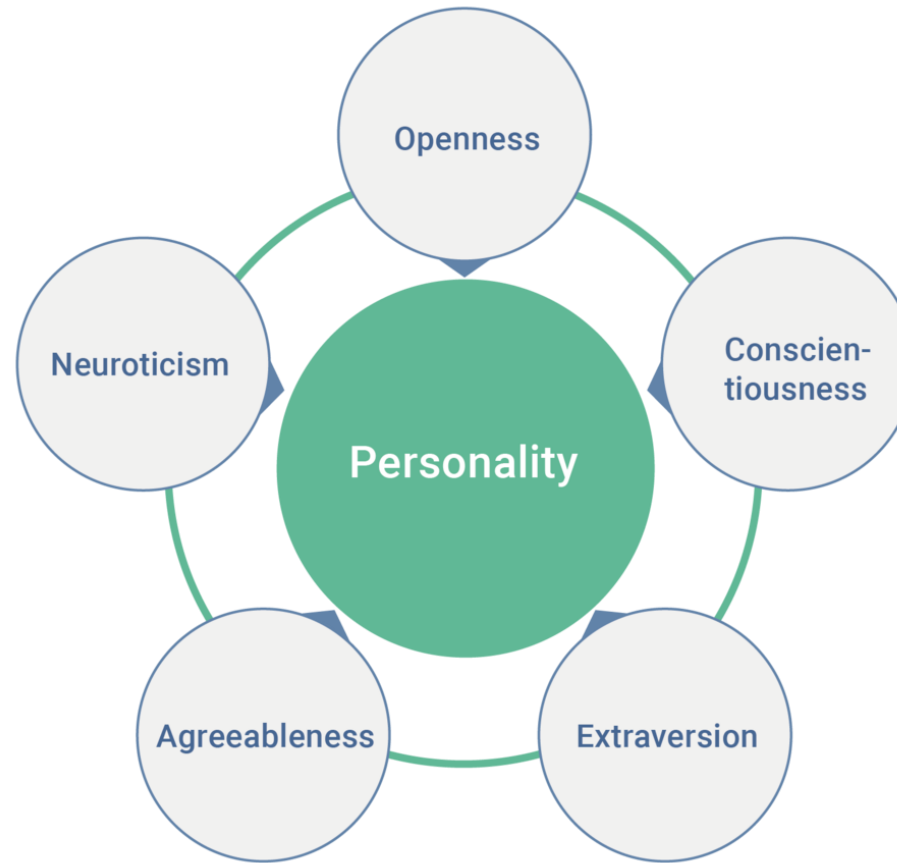
- Electroshock therapy seems to affect the chance of recovery
 - Treatment and recovery are *associated*
 - In this sample → Descriptive statistics
 - In week 9 we will discuss inferential statistics

Today

1. Variability in the data
 - Standard deviation and variance
 - z-score
 - Quartiles
 - Boxplot
2. Associations
 - Between two categorical variables
 - **Between two quantitative variables**
3. Recap
 - Next time
 - Example exam question

Association between two quantitative variables

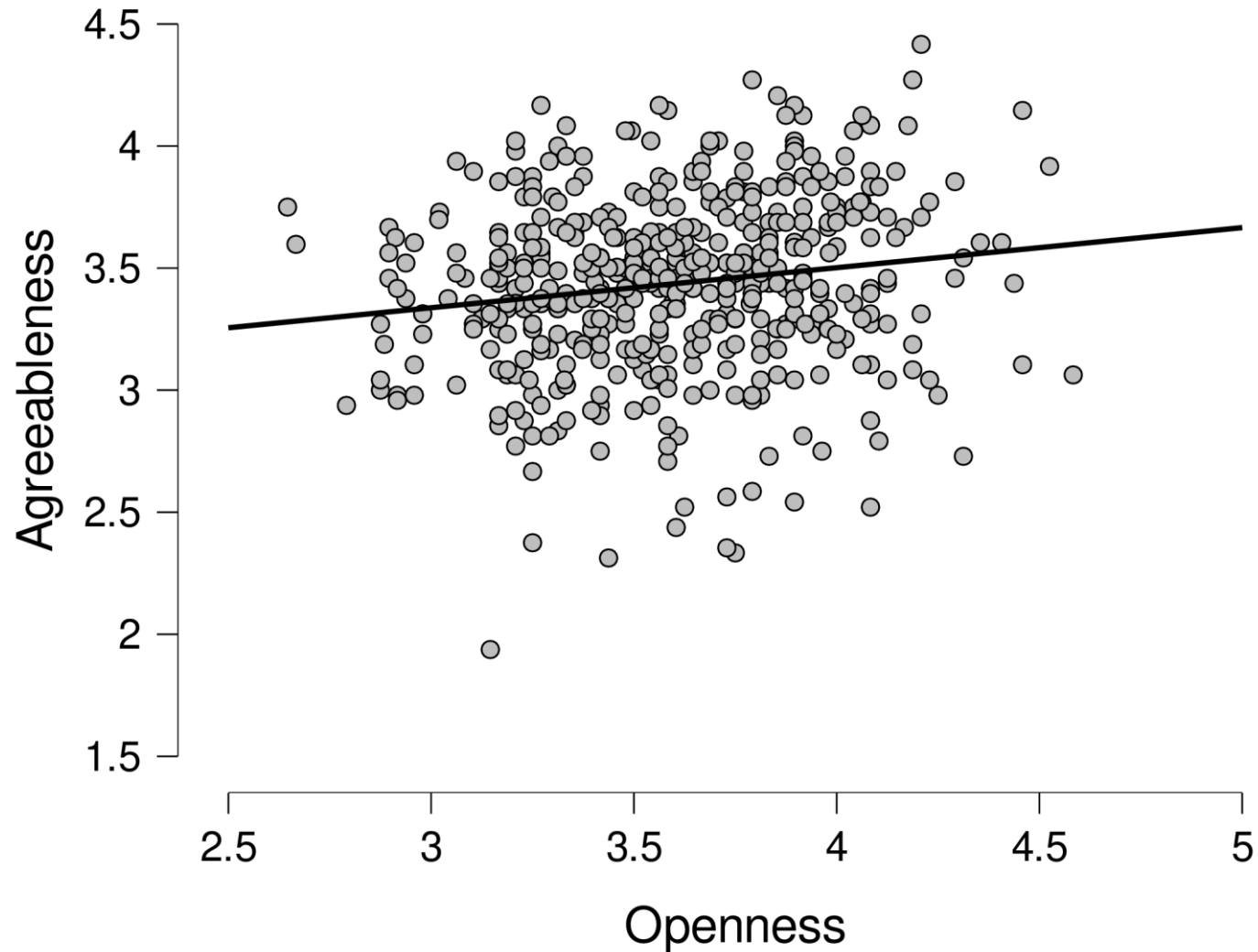
Big 5 Personality Inventory



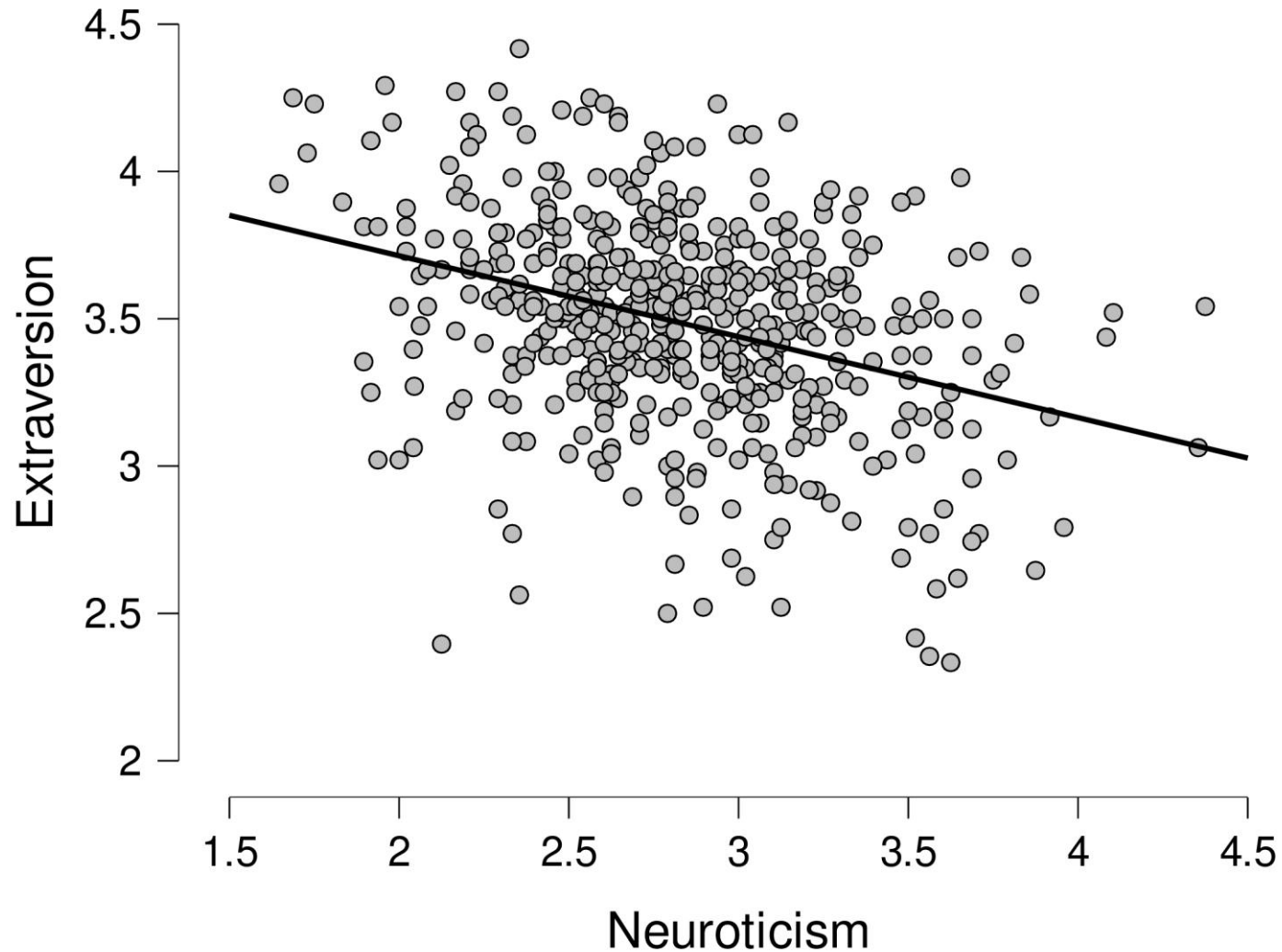
Big 5 Personality Inventory

- Openness
 - “I am quick to understand things.”
- Conscientiousness
 - “I am always prepared.”
- Extraversion
 - “I am the life of the party.”
- Agreeableness
 - “I feel others' emotions.”
- Neuroticism
 - “I get stressed out easily.”

Association – Big 5 Personality Inventory



Association – Big 5 Personality Inventory



Correlation

$$r = \frac{1}{n-1} \sum z_x z_y \quad (\text{Agresti, p. 154})$$

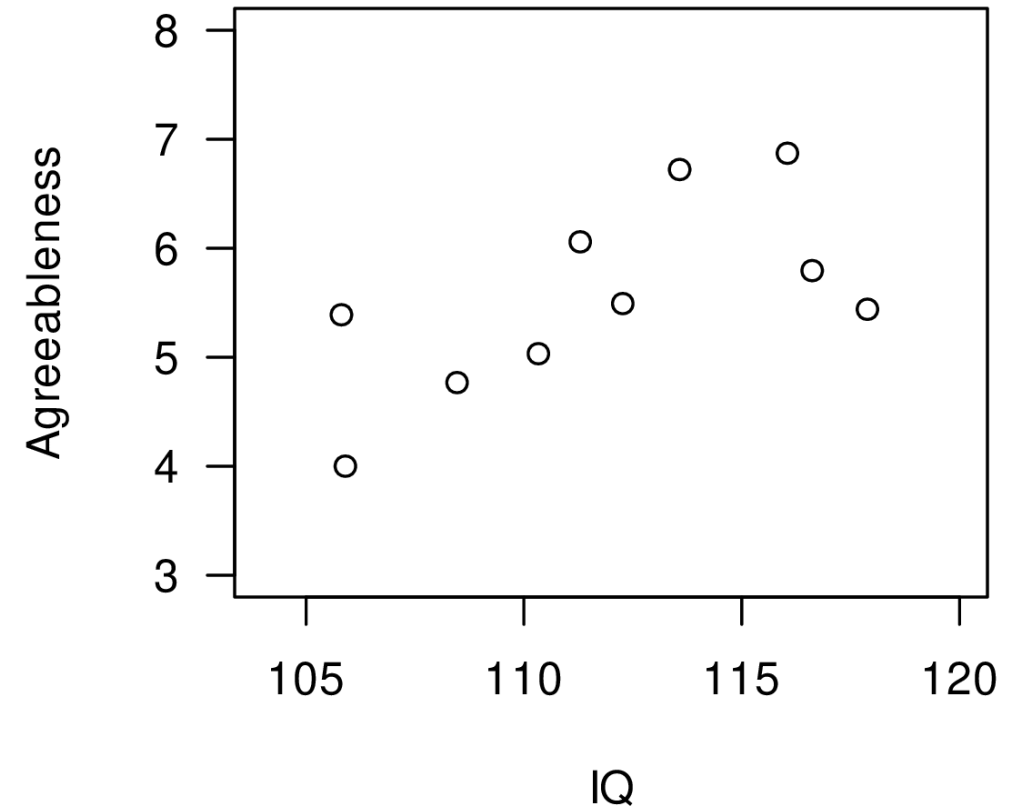
r is “average” contribution to the association of each pair of observations

Properties correlation:

- r between -1 and 1
- $r > 0$: positive association
- $r < 0$: negative association
- Does not depend on the scale of the variables
- Remains the same if the role of response and explanatory variables flips

Correlation

	Agree		IQ	
	6.7		113.6	
	5.4		105.8	
	5.4		117.9	
	4.8		108.5	
	5.8		116.6	
	6.9		116.0	
	5.0		110.3	
	5.5		112.3	
	4.0		105.9	
	6.1		111.3	
Mean	5.6		111.8	
s	0.87		4.3	



Correlation

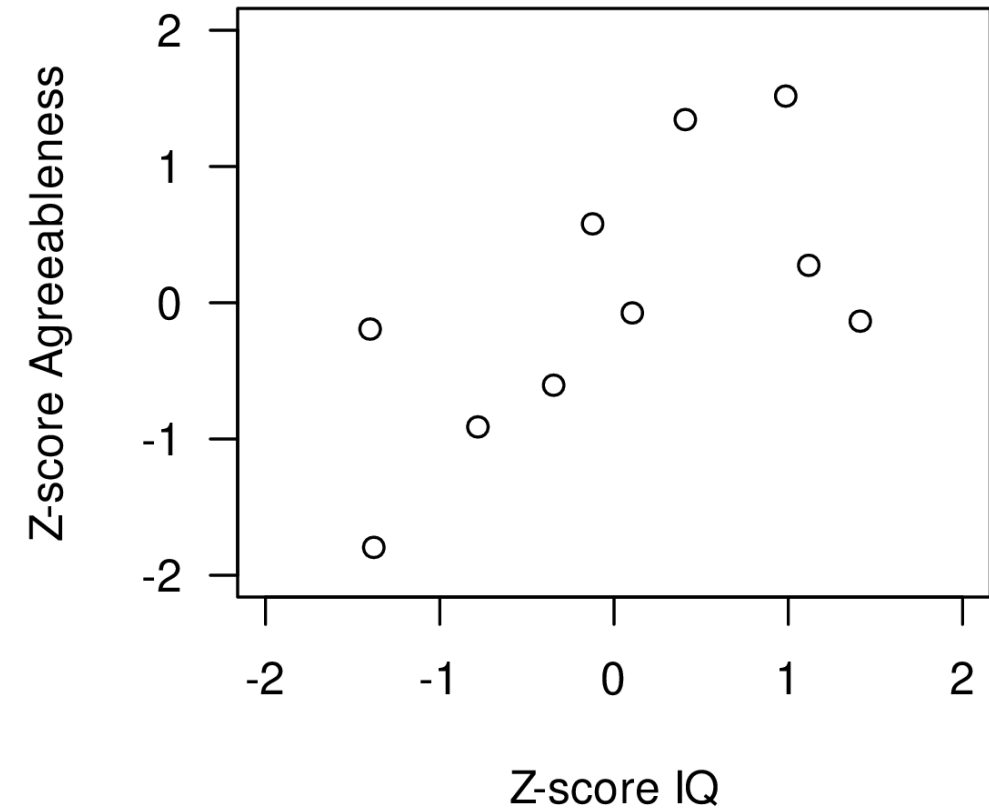
e.g., $(6.7 - 5.6) / 0.87 = 1.3$

	Agree	Z-agree	IQ	
	6.7	1.3	113.6	
	5.4	-0.2	105.8	
	5.4	-0.1	117.9	
	4.8	-0.9	108.5	
	5.8	0.3	116.6	
	6.9	1.5	116.0	
	5.0	-0.6	110.3	
	5.5	-0.1	112.3	
	4.0	-1.8	105.9	
	6.1	0.6	111.3	
Mean	5.6	0	111.8	
s	0.87	1	4.3	

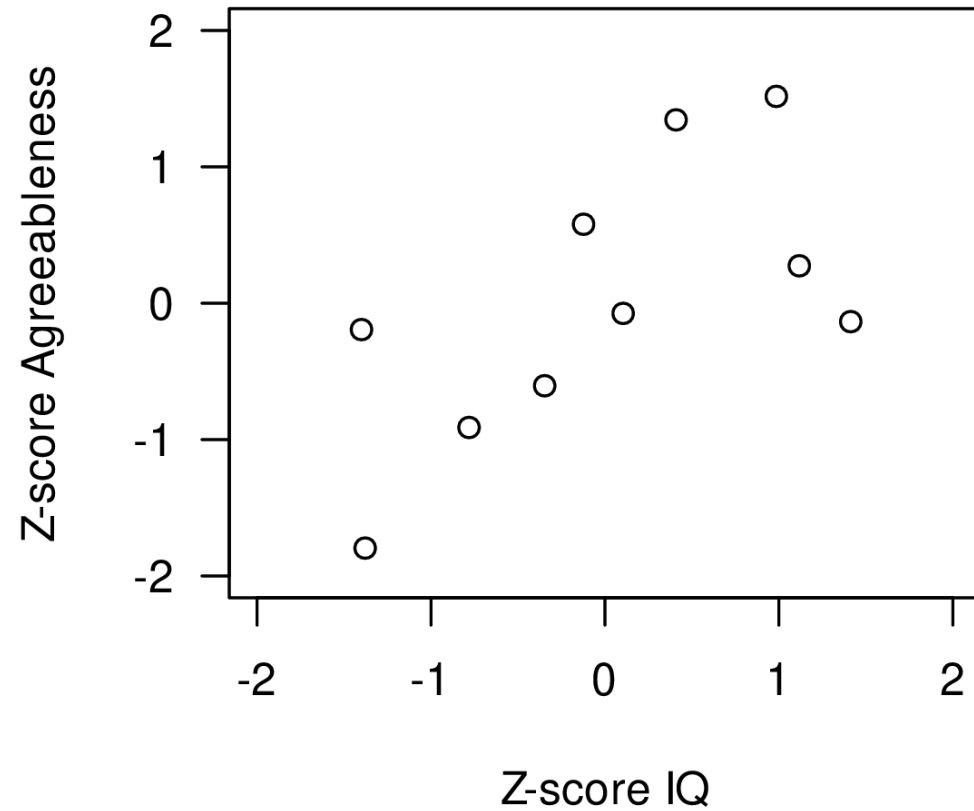
Correlation

e.g., $(113.6 - 111.8) / 4.3 = 0.4$

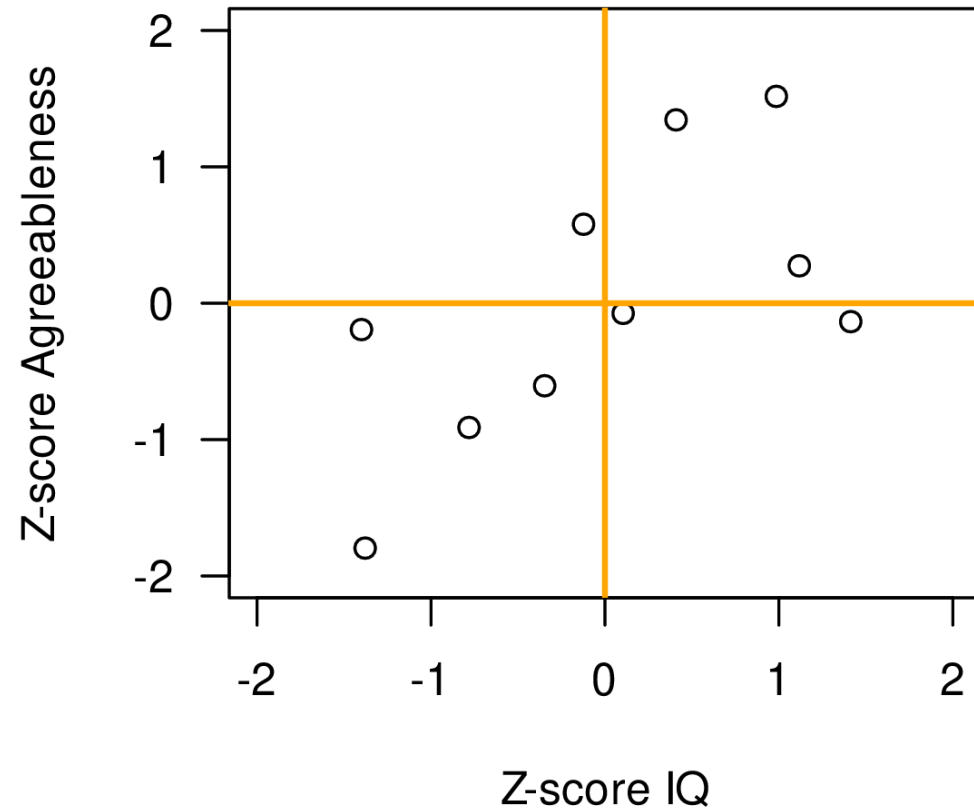
	Agree	Z-agree	IQ	Z-IQ
	6.7	1.3	113.6	0.4
	5.4	-0.2	105.8	-1.4
	5.4	-0.1	117.9	1.4
	4.8	-0.9	108.5	-0.8
	5.8	0.3	116.6	1.1
	6.9	1.5	116.0	1.0
	5.0	-0.6	110.3	-0.3
	5.5	-0.1	112.3	0.1
	4.0	-1.8	105.9	-1.4
	6.1	0.6	111.3	-0.1
Mean	5.6	0	111.8	0
s	0.87	1	4.3	1



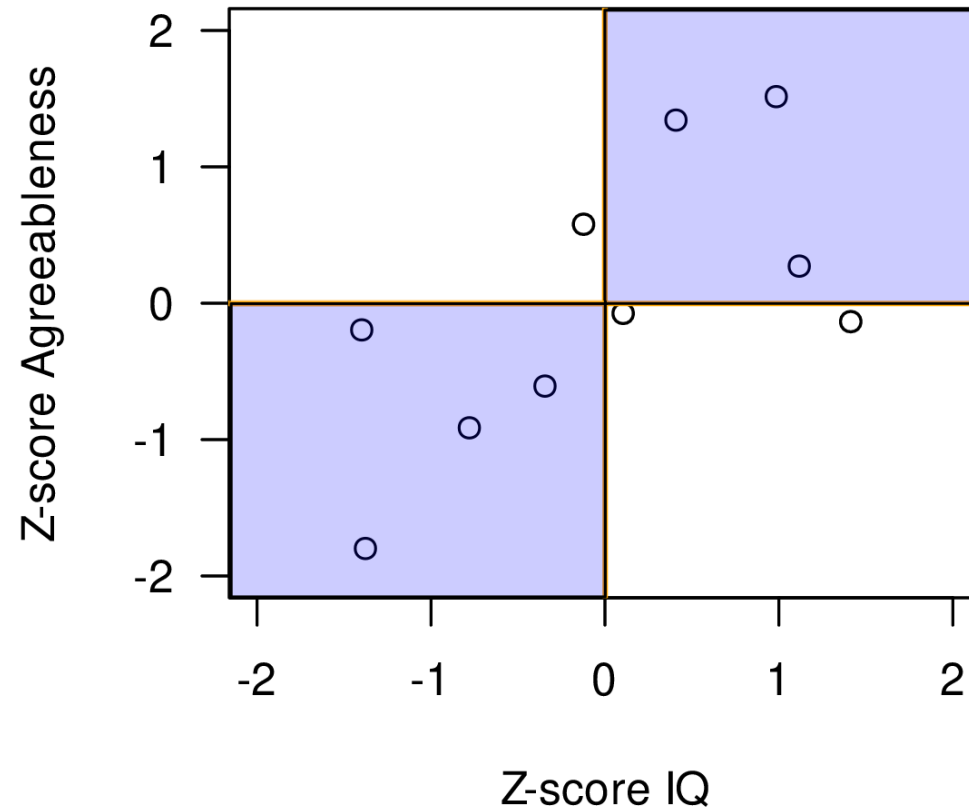
Correlation



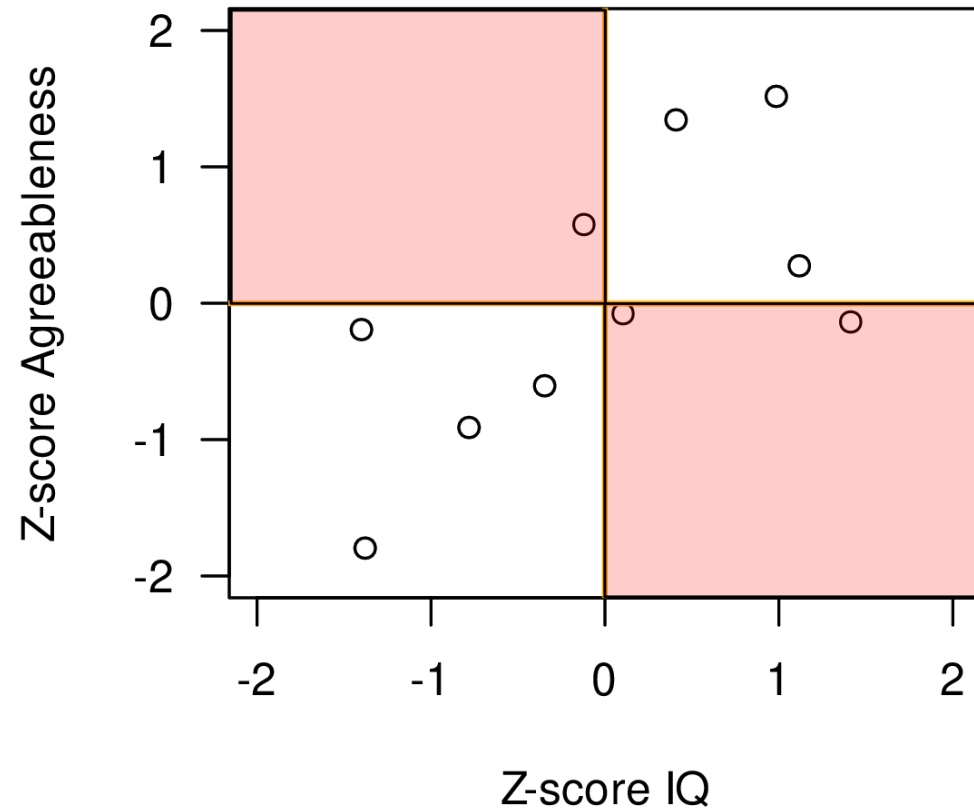
Correlation



Correlation

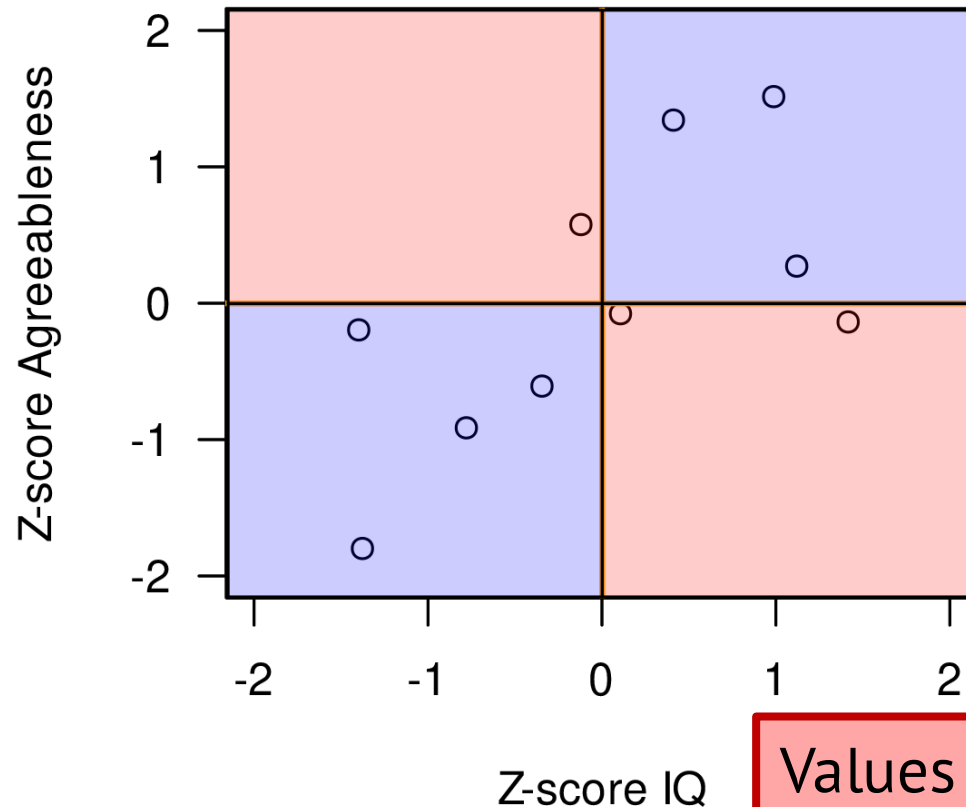


Correlation



Correlation

Values in the blue quadrant: $Z_x Z_y > 0$
Contribute to *positive* correlation

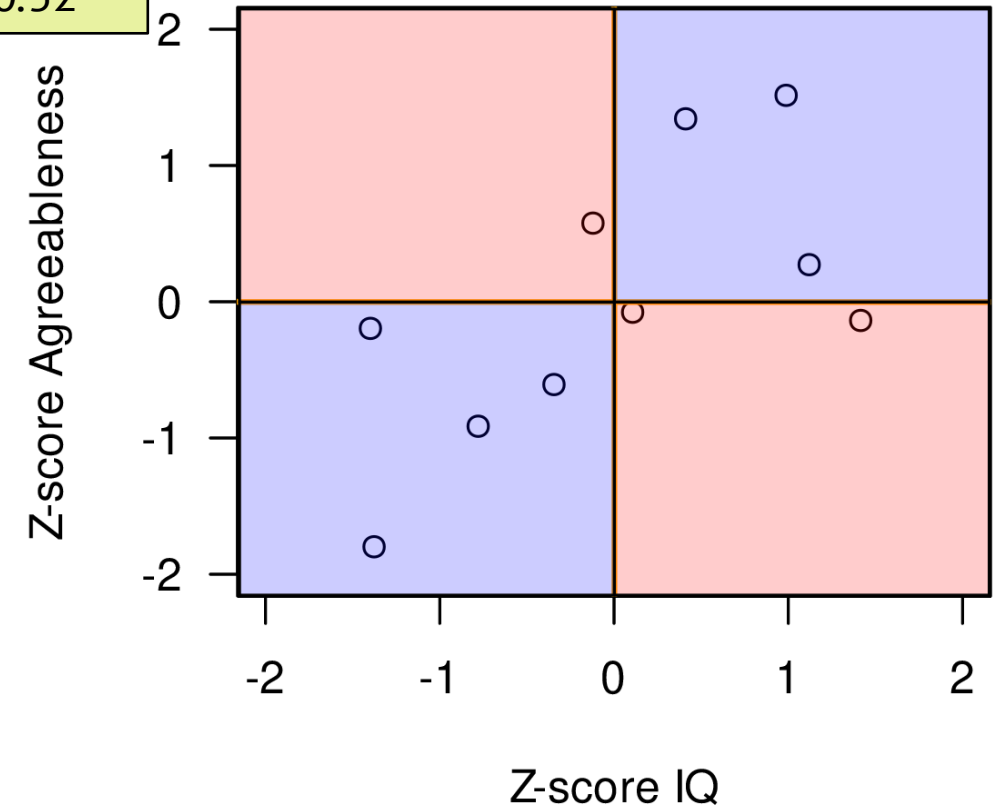


Values in the red quadrant: $Z_x Z_y < 0$
Contribute to *negative* correlation

Correlation

Z-agree	Z-IQ	$Z_x Z_y$
1.3	0.4	0.52
-0.2	-1.4	0.28
-0.1	1.4	-0.14
-0.9	-0.8	0.72
0.3	1.1	0.33
1.5	1.0	1.50
-0.6	-0.3	0.18
-0.1	0.1	-0.01
-1.8	-1.4	2.52
0.6	-0.1	-0.06

e.g., $1.3 * 0.4 = 0.52$



Correlation

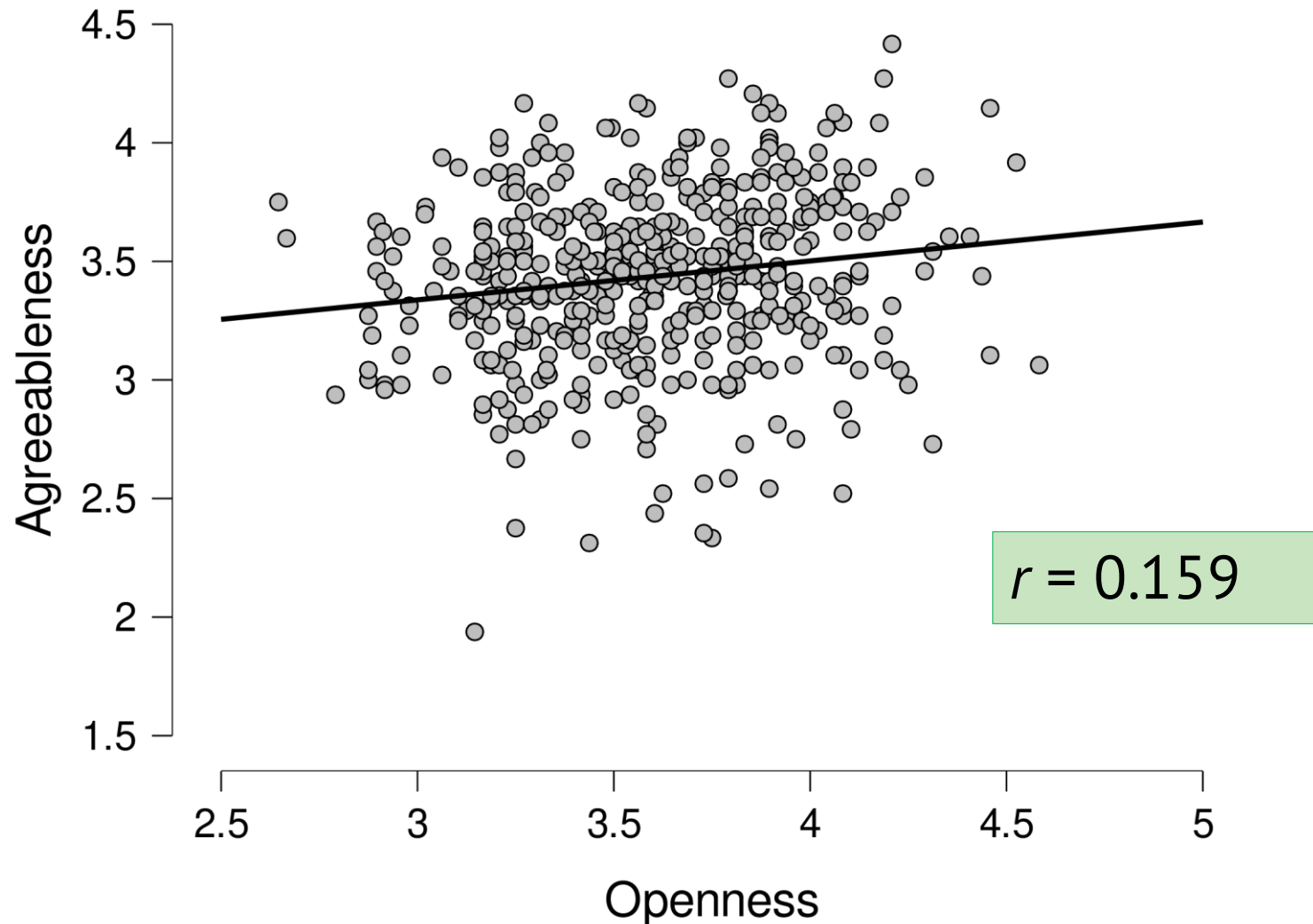
Z-agree	Z-IQ	Z _x Z _y	
1.3	0.4	0.52	
-0.2	-1.4	0.28	
-0.1	1.4	-0.14	
-0.9	-0.8	0.72	
0.3	1.1	0.33	
1.5	1.0	1.50	
-0.6	-0.3	0.18	
-0.1	0.1	-0.01	
-1.8	-1.4	2.52	
0.6	-0.1	-0.06	
		5.84	Sum

$$r = \frac{1}{n-1} \sum z_x z_y$$

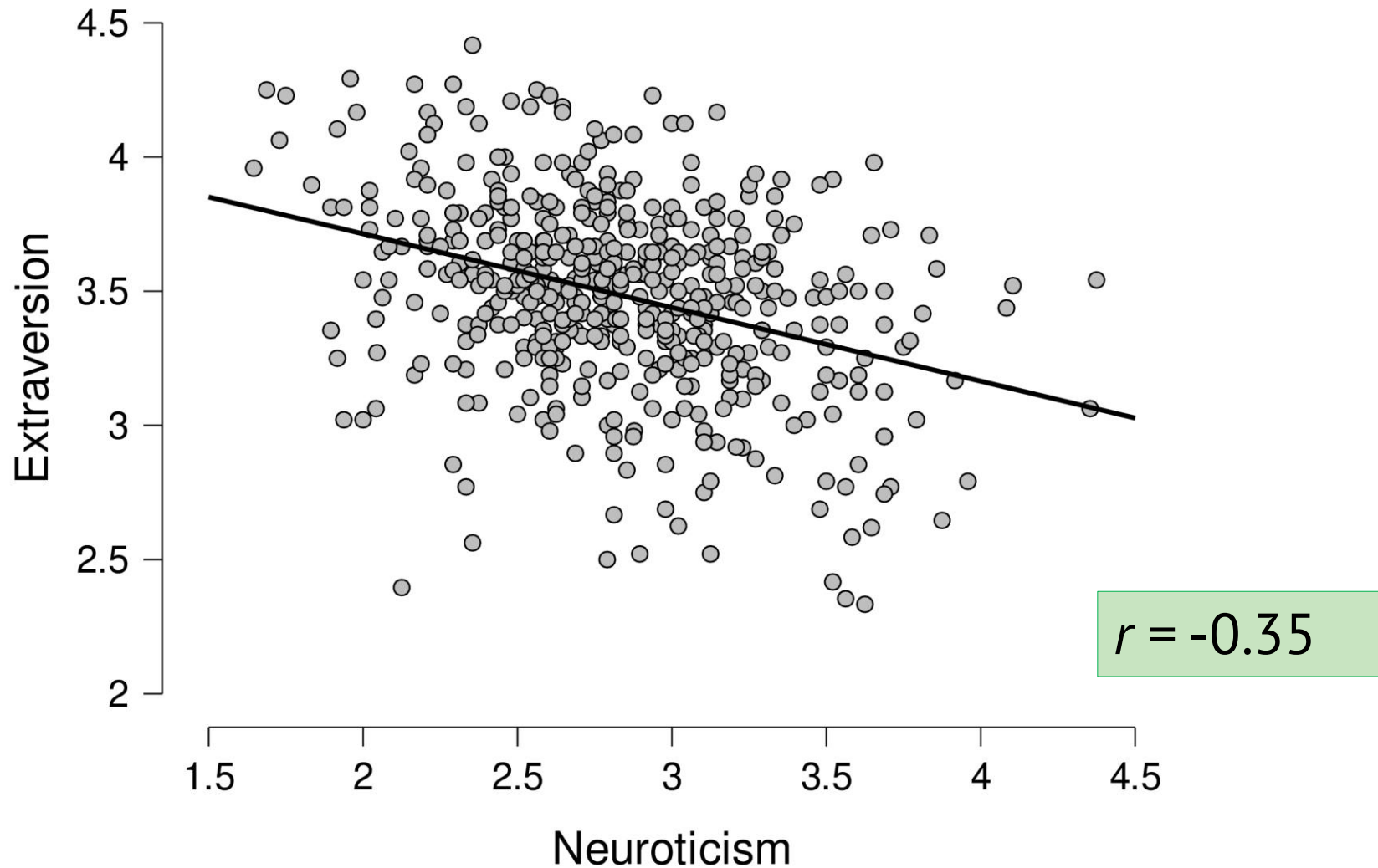
$$n = 10 \quad \sum z_x z_y = 5.84$$

$$r = 5.84 / (10 - 1) = 0.65$$

Association – Big 5 Personality Inventory



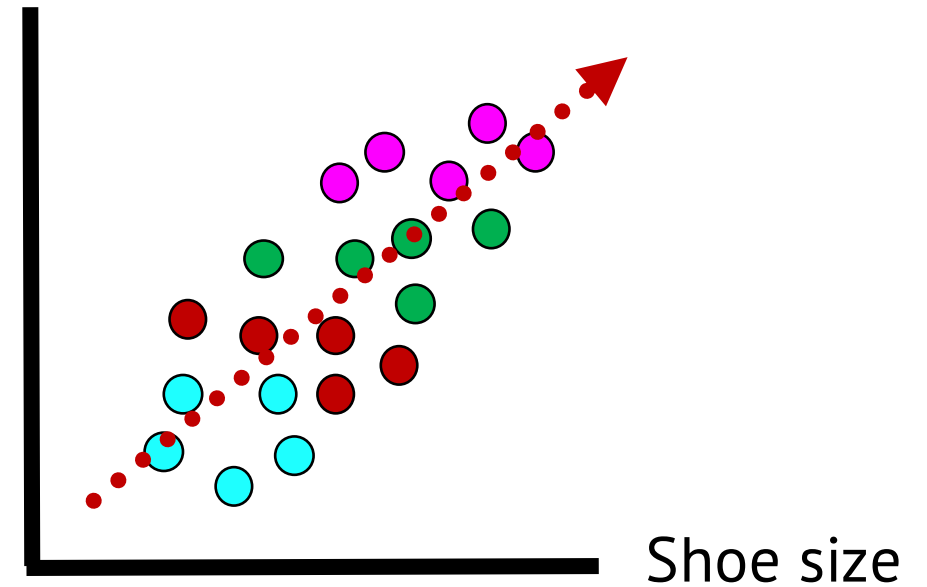
Association – Big 5 Personality Inventory



Wrong inference, beware of confounds!

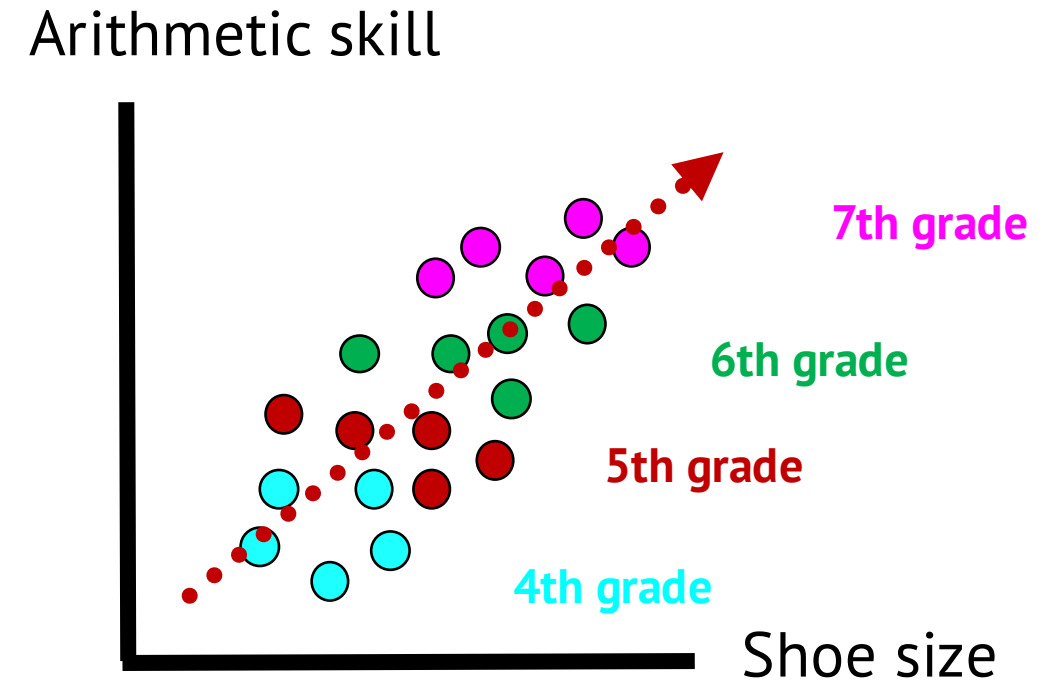
- You can not conclude a causal relationship based on an association
 - More arithmetic skill does not cause larger shoe size
 - Larger shoe size does not cause more arithmetic skill

Arithmetic skill

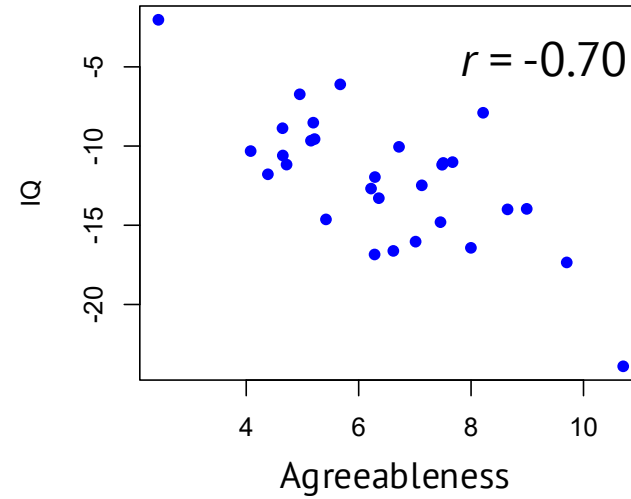
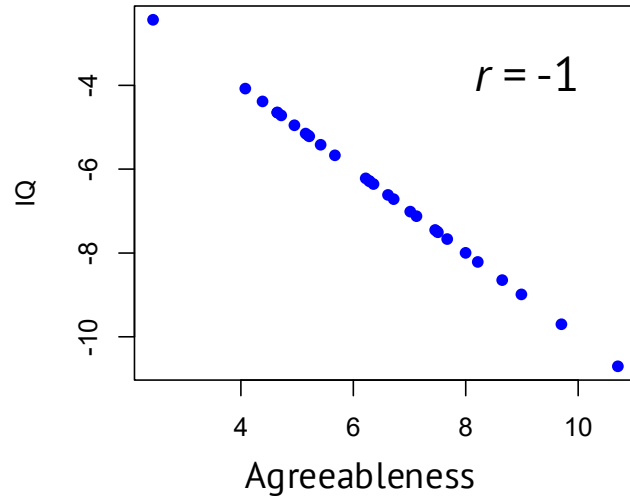


Wrong inference, beware of confounds!

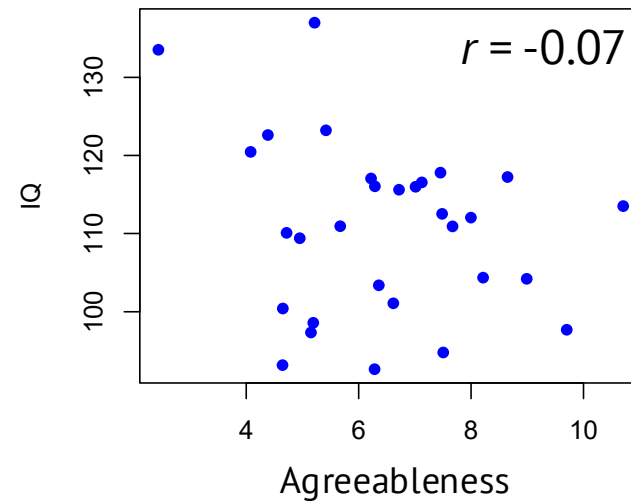
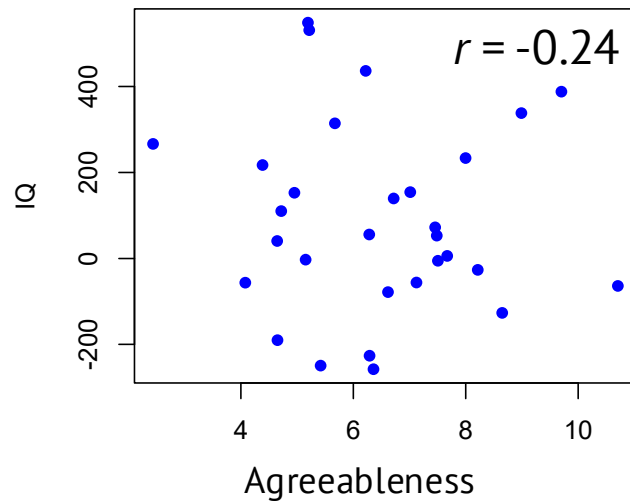
- You can not conclude a causal relationship based on an association
 - More arithmetic skill does not cause larger shoe size
 - Larger shoe size does not cause more arithmetic skill
 - **We only observe an association!**
 - Lecture 8 covers requirements for causal claims



Correlation strength

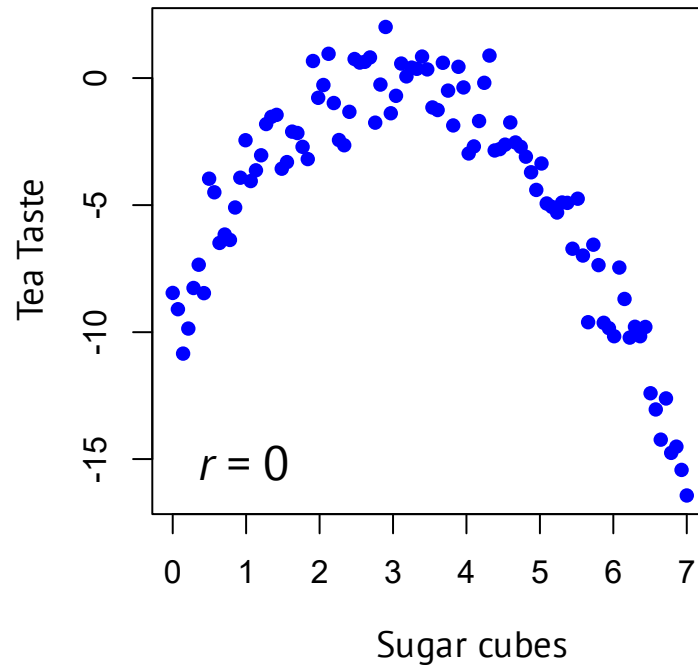


⑦ Strong

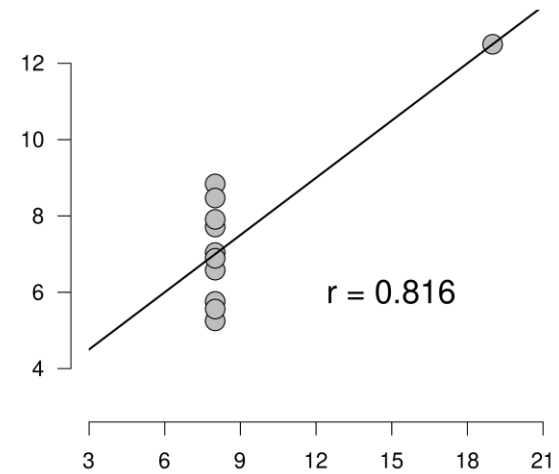
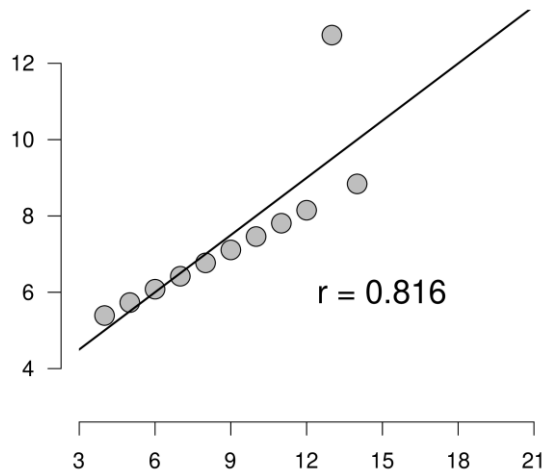
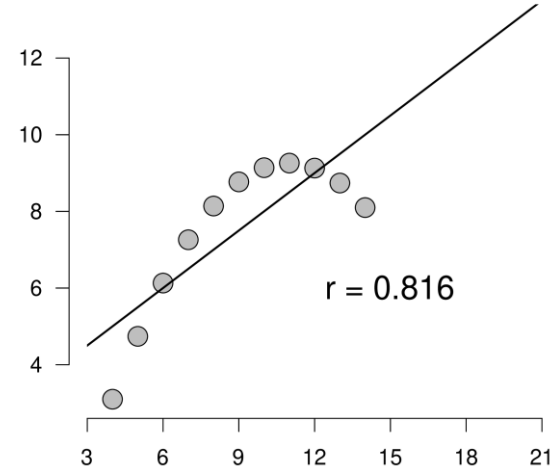
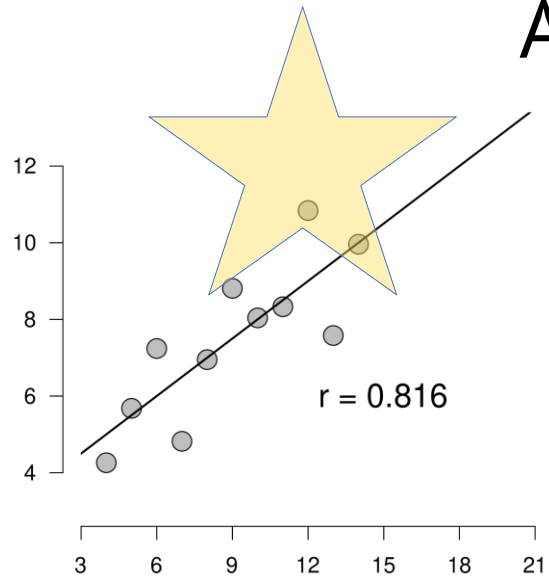


⑦ Weak

Correlation is a linear association



Anscombe's Quartet



Today

1. Variability in the data

- Standard deviation and variance
- z-score
- Quartiles
- Boxplot

2. Associations

- Between two categorical variables
- Between two quantitative variables

3. Recap

- Next time
- Example exam question

Recap of Today

- The distribution of a variable is determined by the mean, shape (skewness), and variability (standard deviation)
- We can standardize variables so we can compare them more easily
- For analyzing the association between two *quantitative* variables, we use the *correlation coefficient*
- For analyzing the association between two *categorical* variables, we use the *contingency table*

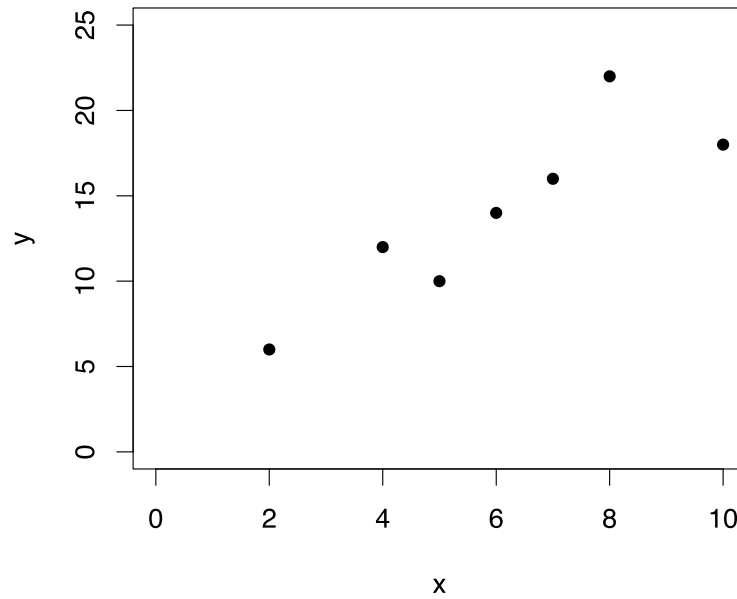
Next time

- Today we talked about variability, proportion etc
- These are concepts from Probability
- To draw inferences, we also need to consider probability
- Next week: reasoning with probability
 - Basic probability rules
 - Conditional probability

Example exam question

- Using the data and the scatter plot below, compute the correlation between x and y

x	y
2	6
4	12
5	10
6	14
7	16
8	22
10	18



a) 0.23

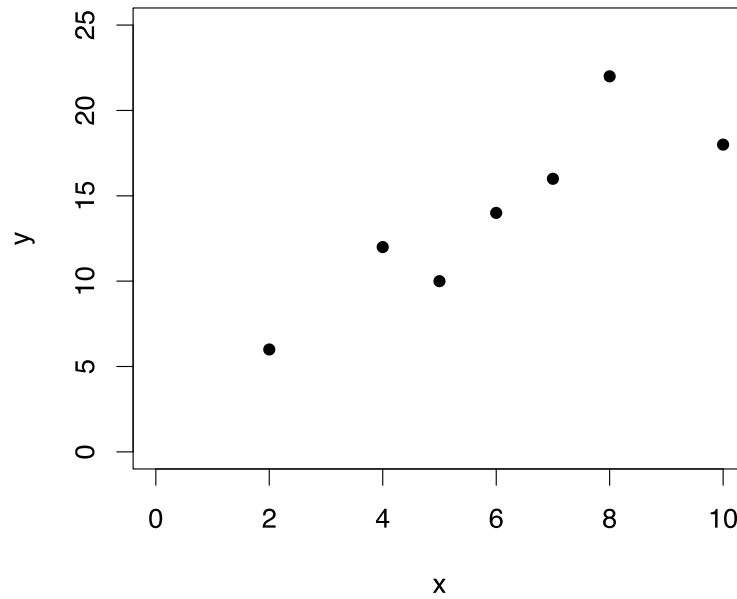
b) 0.76

c) 0.88

Example exam question

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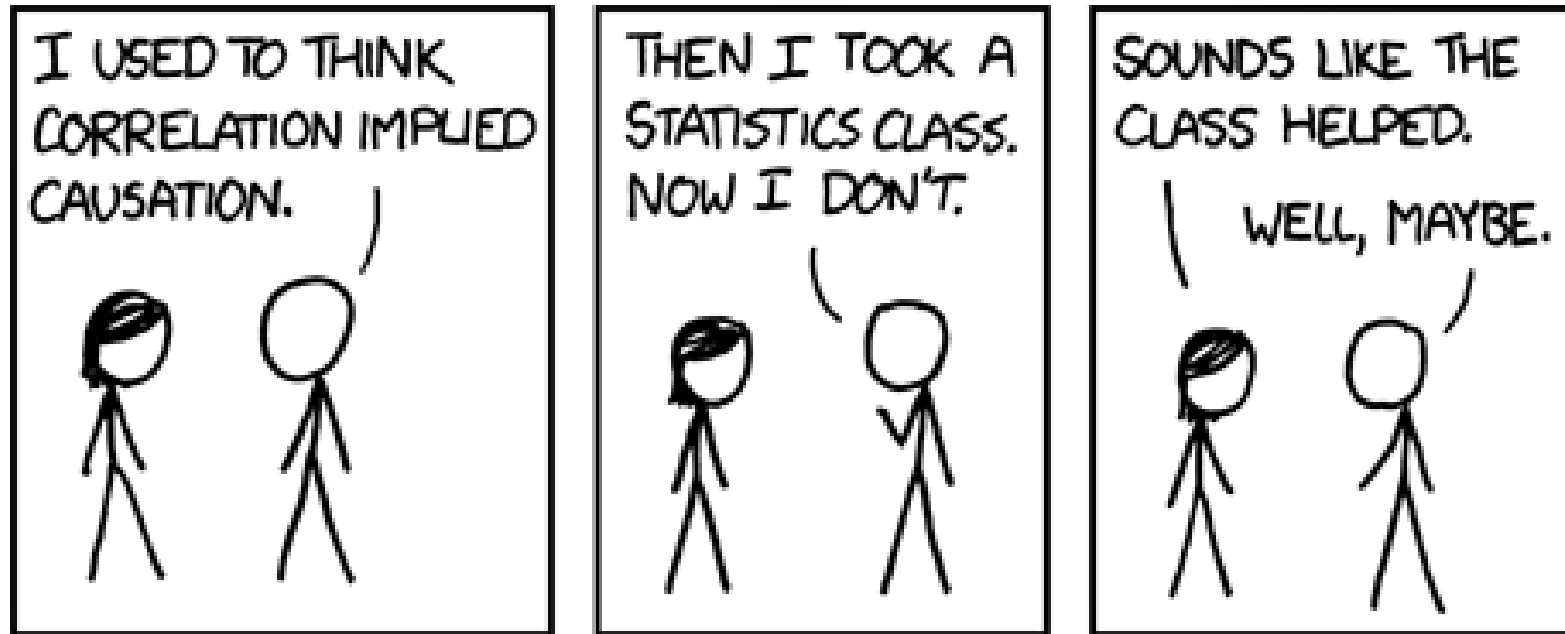
~~a) 0.23~~

~~b) 0.76~~

c) 0.88

Questions?

Thank you for your attention



Bonus Page

Spurious Correlations – Ridiculous things that correlate due to chance or confounding variables (or do they??)

<http://www.tylervigen.com/spurious-correlations>

